

The Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Curated Corpora

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Abstract

When a language model answers from a curated corpus via graph-based retrieval, reported grounding uplift mixes faithful *delivery* of facts already shown to the model with *reasoning* over injected structure, because gold answers derive from the same corpus. We propose *emphexposure* accounting: classify every gold item by whether the shown context exposed it and whether the answer recovered it, and report the four counts beside recall. The

emphcopy ceiling, the recall a verbatim copy of the shown context would achieve, is that table’s per-item scalar and the signed

emphgain over copy its net; both are deterministic, judge-free, and paired with a seed-disjoint placebo. On a production ontology-serving node, mean unaided recall is 0.26 (range 0.151–0.375) and grounded recall about 0.92, yet across ten models from eight developer families gain over copy is uniformly negative (−0.067 to −0.022). The accounting is the finding: of 11,360 gold items, three unexposed items are recovered in total, context utilisation spans 0.900–0.955, and models differ only in how much exposed gold they drop. On items the scaffold does not expose, recovery falls from 0.121 bare to 0.004 grounded: injection displaces parametric recall the bare model demonstrably has. The uplift is *delivery*, measured rather than inferred. A paired study on the same node, varying only the serving path, lifts judged quality by +0.27 pooled (exact $p = 0.002$) and +0.79 where curation is deepest. A uniform-budget control rerun with a length-matched fluent-noise arm shows the scaffold beats no context (+0.75, Holm-significant) but does not establish that the specific retrieved content beats generic on-corpus text. Out of domain the general-set interval (+0.05, [0.00, +0.13]) lies inside a pre-specified ± 0.25 margin (58/60 pairs tie). A two-edge composition test with single-premise controls separates models that compose exposed structure from those that do not; a second local family replicates both findings. Applied to a second domain, typed skill routing over 115 shown rubrics, the scalar moved from +11.6 to +4.7 points under two repairs to one definition of the ceiling while the exposure classification did not move at all; that is the case for reporting the accounting rather than the scalar. Evaluations whose gold derives from the injected corpus should account for exposure as standard, and report the copy ceiling as its summary.

1 Introduction

Enterprises need language models to answer from curated, organisation-controlled corpora that pretraining does not reliably cover. Retrieval-augmented generation bridges parametric memory and required knowledge [16, 30], particularly for long-tail and proprietary facts [37]. Structured, graph-based retrieval further sharpens this approach [3, 10, 20, 43]. Yet when gold answers derive from the injected corpus, headline “grounding uplift” conflates faithful *delivery* of exposed facts with *reasoning* over injected structure. Published knowledge-graph grounding reports lack a standing control that separates them.

We introduce *exposure accounting*, and the *copy ceiling*¹, as its per-item scalar, together with the signed *gain over copy*. Exposure accounting classifies each gold item by whether the shown text exposed it and whether the answer recovered it, and reports the four resulting counts. The copy ceiling is the recall a no-op extractor of exactly the shown text would achieve, which is the exposed fraction; gain over copy is grounded recall minus that ceiling, which pooled over items is the net of two of the four counts. The counts depend only on a surface matcher applied to fixed text, so they carry the findings; the scalar summarises them, and §12 shows how far a summary can move under implementation choices the counts are immune to. This deterministic, judge-free control is paired with a verified seed-disjoint placebo. It generalises the discrete input-only baselines used to interpret reading comprehension [26] and natural-language inference [21, 48]. Grounding evaluations whose gold derives from the injected corpus should report the accounting as standard, with the ceiling as its summary [28, 39].

Without assistance, the evaluated models score 0.265 on average (range 0.151–0.375) on in-domain questions and ≈ 0.92 with curated context. Across ten models from eight developer families, gain over copy is uniformly negative (−0.067 to −0.022), while beyond-exposure recovery totals three items in 11,360 (§8.2). Models therefore differ almost entirely in how much surfaced gold they drop. A clarification on “private”: the corpus is not secret. It is published on GitHub and at a public site (§13), so the non-trivial raw scores (0.151–0.375 across models) reflect partial pretraining

¹The term *copy ceiling* is used independently and in an unrelated sense by Liu [34] (a decoding-layer editing mechanism, where it denotes token reachability under a line-level copy primitive; that paper was withdrawn in 2026), which we flag only to head off a terminology collision.

exposure rather than a genuinely unseen domain; what makes it a useful test bed is that it is organisation-controlled and curated, not that it is hidden. Throughout, “private” denotes the *deployment* pattern (LAN-confined, no cloud egress), not corpus secrecy. High-fidelity delivery of an authoritative, human-vetted source is the product: answers inherit its trustworthiness, are attributable to a named generation, and amortise curation rather than repaying it per query.

Five methods below are corpus-general and separable from the node they are measured on: the copy-ceiling computation and signed gain over copy (§3, §5); the seed-disjoint placebo-verification procedure (§3.4, §8.5); two-edge chain mining with single-premise controls (§11); the blind before/after judged gate for write paths (§10.1); and the finding that absence-keyed fallbacks cannot detect wrong-but-present retrieval (§6). The harness (paired bootstrap, signed-rank tests, matched-pairs rank-biserial, Holm), the frozen question sets, the per-row sweep and Study-2 data, and the served node are released in the repositories footnoted below; per-study bootstrap seeds are named in the released scripts (7 for the legacy Study-2 analysis, 42 for the decomposition and re-score), artefacts for the newest studies (control rerun, composition, replication) are being consolidated into the same repository and any not yet present at the tagged commit should be treated as reported-but-not-yet-released.

2 Related Work

Knowledge-graph and private-corpus grounding. Grounding lineage runs from RAG [30] through graph-structured retrieval [10, 20] and zero-shot knowledge-graph triple prompting [3] to roadmaps and surveys [43, 47]. GraphRAG addresses “private and/or previously unseen” collections [10], while enterprise variants highlight amortised curation [38]. Published work typically injects *instance* triples or opaque encodings, including community summaries [10] and GNN soft-prompts over retrieved subgraphs [22]. We instead inject human-readable, schema-level content: class definitions, typed relations, and taxonomy. None of these graph-grounding studies reports a standing input-exposure control beside its uplift figures (claim-level instruments that measure related quantities are compared in §3.5), so their headline uplift is not re-centred on content already present in context — though where retrieval coverage is reported, a reader can approximate the subtraction from their published quantities.

Baselines that reframe a result. Simple controls can overturn headline results: strong RAG baselines match elaborate pipelines under equal token budgets [28]; controlled comparisons reverse earlier “RAG wins” findings and add a confidence router [32]; and apparent argument comprehension can reflect spurious cues [42]. The confidence router of Li et al. [32] and selective retrieval [2, 24, 37] provide precedents for our injection gate. Those methods choose whether to

retrieve; ours chooses whether to inject.

Input-only baselines and faithfulness metrics.

Our copy ceiling extends this tradition of input-only baselines. Passage-only, question-only [26], and hypothesis-only baselines [21, 48] reveal shortcuts present in inputs. Unlike these discrete task classifiers, the copy ceiling is continuous and per-item. Faithfulness can be measured separately from correctness [1, 12, 64]; metric-first evaluations formalise assumptions [39] and test shortcut robustness [15]. FActScore’s source precision and Adlakha’s token-overlap K-Precision resemble our exposure count. ALCE rejects “copy the passage” as a shortcut; we use it as the reference point. Contamination audits [18, 51] assess input-visible score post hoc, whereas we define deliberate exposure as the reference a priori. The closest ancestry is the extractive-oracle and lead-*n* baselines of neural summarisation [40, 53] and the trivial-baseline critiques of reading comprehension [26], which likewise ask what a copy of the input already achieves. A literature search (2026-08-24; arXiv and ACL Anthology, grounding/faithfulness/RAG-evaluation queries) found no direct precedent for a judge-free, per-item copy baseline reported as a standing control for knowledge-graph-grounded generation; RAGChecker’s ground-truth claim recall in retrieved context [50] is the closest comparator (§3.5), and the contribution claimed is the deterministic instrument and reporting convention, not priority over the input-copy idea.

Nearest neighbours, and how we differ. Closest in method is work that causally manipulates the gold answer to test its effect on retrieval-augmented gains [31]; we adopt only its paired-intervention *method* (the authors withdrew the work in 2026 after finding errors that compromised its main conclusions). Leakage-free benchmarks instead extract reasoning graphs from question–context pairs, then apply type-constrained entity replacement with consistency checks and leakage filters [33]; this occurs during corpus construction, whereas the copy ceiling is a per-item measurement. RAGChecker uses LLM extraction and grading to decompose RAG quality into claim-level retrieval and generation scores [50]. Our context utilisation is a deterministic, surface-matched version of its context-utilisation metric; we claim no originality for the decomposition. MIRAGE measures benchmark-level adaptability by evaluating reader behaviour against oracle context across retriever–model pairings [46]. The RAGAS *library*’s later “context recall”, absent from the original paper [12], is an LLM-judged retrieval-coverage measure. We have not found elsewhere our combination of a deterministic, judge-free, continuous per-item exposure ceiling for a schema-level ontology corpus and a verified seed-disjoint placebo. Across models, signed gain over copy provides an exposure-normalised recall scale where raw uplift does not distinguish models. Judged arms use a cross-family judge to limit self-preference [45, 63].

3 The Copy Ceiling

We treat exposure accounting as the paper’s central contribution and the copy ceiling as its reportable scalar, and we therefore provide a formal definition of both, an explicit list of assumptions, and a dedicated robustness design, in line with the metric-first precedent [15, 39]. The order matters: the scalar is what a reader will quote, but the accounting is what survives a change of implementation, and §12 measures the difference on our own instrument.

3.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and answer a_i . Let $m(t, X) \in \{0, 1\}$ be the deterministic surface matcher, equal to one when the normalised title of gold item t , or at least 80% of its length- ≥ 4 words, appears in X . Define the exposed count and per-item **copy ceiling**

$$e_i = \sum_{t \in G_i} m(t, S_i), \quad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = (\sum_{t \in G_i} m(t, a_i)) / |G_i|$. A no-op extractor returning S_i verbatim scores exactly c_i . Exposure is computed over the full visible input (the injected scaffold plus the question text); on the frozen sets the question text adds no exposure beyond the scaffold on any of the 510 questions, and for the raw arm (question text only) exposure is small but non-zero: 35 of 1,136 gold items appear in the bare question, a mean raw-arm ceiling of 0.037.

Scoring symmetry, stated and checked. One question template (T-COMMON, 45 of 510 questions) accepts *any* of several valid targets; the headline answer scorer credits such a question fully when any target is matched, and the released implementation historically computed the ceiling as the exposed *fraction* for these questions while collapsing the answer score. On the frozen sets this asymmetry is numerically inert: every multi-target question’s alternatives are exposed all-or-nothing, so the symmetric scorer (the same question-level scoring function applied to the answer and to the copy text) reproduces every reported gain to four decimal places, and the executable invariant — the concatenated visible input, scored as an answer, equals the reported per-item ceiling — holds on all 510 questions under both definitions. The released `decompose_exposure.py` now runs this invariant as a hard check.

Assumptions (when the instrument applies).

- (A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so gold is deliberately present in the injected text. The ceiling measures this exposure; it is not a leak.
- (A2) **Shared matcher.** c_i and r_i use the same matcher. The matcher is a surface proxy: it

under-counts paraphrase, and it can also over-count (its exact-title path is a substring test, its long-word threshold ignores word order, relation roles and negation, so naming a correct entity in a wrong or negated relation still scores). Absolute recall is therefore neither a strict lower nor upper bound on semantic correctness; the paired difference $r_i - c_i$ is first-order insensitive to *shared* matcher error, but cancellation is approximate because error rates may differ between structured scaffold text and generated prose. Where paraphrase is a live threat we re-score with a semantic matcher (§4).

- (A3) **Per-item, model-free ceiling.** c_i depends only on (q_i, S_i) and is identical across models given a fixed scaffold, enabling comparison across a model sweep.
- (A4) **Graph-derived gold.** The ceiling is informative only when gold derives from the injected corpus. Where gold is independently authored (the out-of-domain arm), $c_i \approx 0$ and judged quality carries the axis.

A recall reference point, not an endorsement. Although the no-op extractor scores c_i , it is not an admissible answer: returning the entire scaffold lacks the required precision, conciseness and relevance. The ceiling is therefore a recall reference for delivery fidelity, not a recommendation to reproduce the context.

3.2 Exposure accounting: the counts are primary

For each gold item $t \in G_i$ write $x_t = m(t, S_i)$ for *exposed* and $y_t = m(t, a_i)$ for *recovered*. Pooling over all gold items of a run gives four counts,

$$\begin{aligned} n_{11} &= \#\{x_t=1, y_t=1\}, & n_{10} &= \#\{x_t=1, y_t=0\}, \\ n_{01} &= \#\{x_t=0, y_t=1\}, & n_{00} &= \#\{x_t=0, y_t=0\}, \end{aligned}$$

with $N = n_{11} + n_{10} + n_{01} + n_{00}$. Every quantity this paper reports is a function of them. The pooled ceiling is the exposed fraction $(n_{11} + n_{10})/N$, pooled recall is $(n_{11} + n_{01})/N$, and their difference, the pooled gain over copy, is

$$\bar{g} = \frac{n_{01} - n_{10}}{N},$$

the net of two cells. Context utilisation, $n_{11}/(n_{11} + n_{10})$, is the fraction of exposed gold that survives delivery; beyond-exposure recovery, $n_{01}/(n_{01} + n_{00})$, is the fraction of unexposed gold the model supplies itself. The headline figures in §8 average per-question ratios rather than pooling items, so they differ from these identities in the second decimal; the released decomposition script reports both.

Two properties make the counts the right object to report. They are diagnostic where the scalar is not: a gain of zero is consistent with $n_{01} = n_{10} = 0$ (perfect delivery, no recovery) and with $n_{01} = n_{10} = 200$ (a model that drops as much exposed gold as it invents),

and only the counts tell those apart. And they are stable where the scalar is not. Each x_t depends on the matcher and on fixed text, and on nothing else: no ranking, no threshold, no tuning. A scalar ceiling for a task without a natural no-op extractor must be *constructed*, and every construction choice is a place it can move. On the routing task of §12 one definition of the ceiling, implemented three ways, gave gains of +11.6, +9.3 and +4.7 points; the classification of which rubric was shown did not change under any of them. The ceiling is the number to quote; the counts are the number to trust.

3.3 Gain over copy

We define the signed **gain over copy** as $g_i = r_i - c_i$, reporting the headline figure $\bar{g} = \bar{r} - \bar{c}$. It re-centres grounded recall against the exposure reference, and pooled over items it is $(n_{01} - n_{10})/N$ (§3.2).

- $g \approx 0$, *delivery fidelity*: the model restates nearly everything shown. Recall measures omission rather than fabrication, leaving additions beyond exposed gold unmeasured; for curated-corpus grounding, restatement is the target behaviour.
- $g < 0$, *lossy restatement*: the model drops surfaced gold. The decomposition of §8.2 measures this directly: the negative gain is almost entirely exposed-but-omitted gold (n_{10}), with unexposed recovery (n_{01}) negligible. The magnitude places models on an exposure-normalised scale that raw uplift, which conflates exposure with delivery, does not provide (with a constant ceiling it preserves the recall ordering; §8.2).
- $g > 0$ would evidence *reasoning over structure*, such as composing an implied but unstated relation. We observe essentially none in-domain ($n_{01} = 3$ of 11,360; §8.2). A positive g alone would not prove reasoning: parametric leakage, guessing or matcher asymmetry could also produce it, and only the multi-hop design of §19 could separate them.

A near-ceiling result shows the measured recall is achievable by delivery alone: the metric cannot credit reasoning beyond exposure, and no reasoning is needed to explain the score. It does not measure how the model selected the correct names internally, so it constrains what the evaluation can attribute, not what the model did. The ceiling is likewise a reference baseline rather than a mathematical upper bound on all valid answers. Figure 1 illustrates the two paths compared.

3.4 Negative controls: what would move the gain

A positive delivery result might reflect lexical overlap rather than knowledge transfer. Four arms hold the model and question fixed while perturbing only the injected block (defined here; results in §8.5):

true the block actually served by the node, injected into the bare model. It should reproduce the live

loom arm, showing that the serving path adds nothing beyond the block.

shuffled the same block with body sentences shuffled while retaining the wrapper and seed. Because whole-sentence shuffling preserves every within-sentence fact, this arm tests sensitivity to sentence *order* only; survival is consistent with order-independence, not proof that structure is unused.

masked the same block with every seed-class title replaced by **Entity- k** , preserving structure but removing seed entity names. Because answer-bearing *target* names remain, this arm cannot test removal of answer-name matching; it probes dependence on seed naming only.

irrelevant the well-formed, on-corpus scaffold of a *different* question, using a fixed derangement over engaged items. Seed-IRI disjointness is a construction check: it prevents donor and target sharing seed classes, but does not by itself guarantee the donor block exposes none of the target’s gold, ancestors or synonyms. Survival would show that any ontology-shaped text suffices.

If gain over copy remains near zero under **shuffled**, **masked** and **irrelevant**, the ceiling tracks exposure and delivery is content-bound; a shift indicates leakage through surface form or position. This experiment tests whether the observed lift is content-specific, serving as the direct analogue of a shortcut-robustness check for a faithfulness metric [15].

3.5 Relation to prior instruments

Several recent instruments ask a related question (did the shown context deliver the answer?) by different means, and the copy ceiling should be read against them. A reader–context diagnostic scores whether a budget-constrained reader recovers the answer from its retrieved context and reports that this predicts downstream F1 above retrieval recall [4]; *Sufficient Context* labels whether the context alone suffices, using a prompted autorater at roughly 93% agreement [25]; RAGChecker decomposes quality into claim-level retrieval and generation scores through LLM extraction and entailment [50]; and an information-bottleneck view filters retrieved noise by compressing context against the answer [65]. The idea of an input-side reference is older still: the extractive-oracle tradition in summarisation (lead and sentence-level oracles [40, 53] and syntactic-compression oracles [59]) makes a copy of the input a near-definitional upper bound on any recall- or ROUGE-style metric.

Against this lineage we claim four specifics, not the input-copy idea itself. First, the ceiling is a *continuous, per-item, verbatim-copy recall*, not a binary answer-survival label or a claim-level score. Second, we report a *signed* gain over copy, which re-centres recall on exposure and can be negative: the diagnostically informative case these instruments do not surface. Third, the ceiling is *judge-free and deterministic*: it needs no autorater and is identical across a model

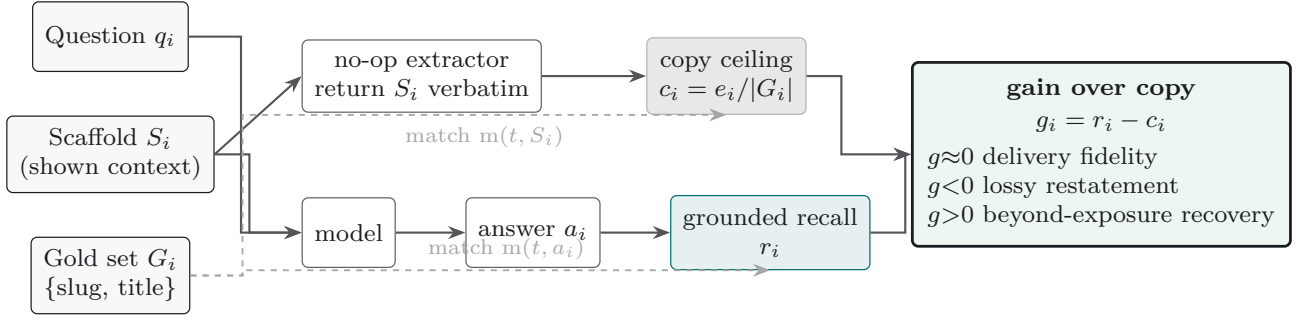


Figure 1: The copy ceiling and the signed gain over copy on one panel. The same shown context S_i feeds two extractors: a no-op extractor that returns S_i verbatim, whose recall against the gold set G_i is the per-item *copy ceiling* c_i ; and the model, whose answer a_i scores the grounded recall r_i . Both scores use the identical surface matcher $m(t, \cdot)$ with G_i as reference (dashed), so their difference, the *gain over copy* $g_i = r_i - c_i$, is first-order insensitive to matcher error the two scores share (cancellation is approximate; error rates can differ between scaffold text and generated prose). The sign reads directly: $g \approx 0$ is faithful delivery (the target behaviour for private-knowledge grounding), $g < 0$ is lossy restatement of surfaced gold, and $g > 0$ would evidence recovery of gold beyond what the exposure check finds (§3).

sweep, whereas the sufficient-context and claim-level instruments depend on a prompted grader. Fourth, we pair it with a *negative-control and composition apparatus*, namely the verified seed-disjoint placebo (§3.4) and the two-edge design (§11), that tests for content-specific transfer and locates the synthesis boundary. The reader-context diagnostic is the closest neighbour and also the clearest caution: it finds verbatim survival only *moderately* predictive of F1. We therefore position the copy ceiling as a calibrated instrument for delivery fidelity, not as a classifier of downstream correctness.

4 Semantic Re-Score: Is the Ceiling a Lexical Artefact?

Because the matcher is lexical, a sceptic can ask whether the negative gain is an artefact of strict matching: the answer paraphrases gold that the ceiling, computed on the same matcher, happens to catch. We re-scored every saved sweep row with a paraphrase-tolerant embedding matcher, re-scoring the copy ceiling under the identical matcher so the comparison stays symmetric. The matcher embeds candidate spans (word 1–6-grams within delimiter-bounded chunks, plus each chunk and the whole text) with `bge-small-en-v1.5` and scores a gold item as recovered when the maximum cosine against its title meets a threshold; we sweep thresholds 0.80/0.85/0.90 and use the paper’s seeded 10,000-resample paired bootstrap. The result is threshold-dependent in level but not in sign: at 0.80 and 0.85 semantic scoring raises both recall and ceiling (ceiling 0.980 and 0.968 against 0.964 lexical); at the strictest threshold, 0.90, both fall below their lexical values. In all thirty model $\times$ threshold combinations the gain over copy remains negative with the 95% CI excluding zero, and pooled beyond-exposure recovery stays in low single digits at every threshold (3 of 11,360 at 0.85; the largest single-model count at any threshold is 4). Per-model gain magnitudes move slightly (Qwen-2.5-72B improves from -0.050 lexical

to -0.034 at 0.80, the largest shift), but no model’s interval crosses zero. The headline is therefore not an artefact of lexical matching. Unlike the lexical instrument, this check depends on an embedding model and its span extraction; it is a robustness probe, not a replacement scorer.

5 Making the Copy Ceiling Runnable

This control is cheap to adopt because it needs no model. To compute it on any corpus:

- Inputs.** A list of questions, each with a gold target set (the $\{\text{slug}, \text{title}\}$ pairs the source asserts as its answer) and the exact context string S_i that would be shown to the model. Gold and context must derive from the same source for the ceiling to be meaningful (assumption A1).
- Matcher.** A deterministic surface matcher $m(t, X)$: one when the normalised title of gold item t , or $\geq 80\%$ of its length- ≥ 4 words, appears in X . The reference implementation and the scaffold engine used here are released in the harness (`tools/paper/decompose_exposure.py` and `ontology_scaffold_v1.py`); a paraphrase-tolerant embedding variant is a drop-in (§4).
- Compute, no model call.** Per question, the copy ceiling is $c_i = (\sum_{t \in G_i} m(t, S_i)) / |G_i|$: the fraction of gold already exposed in the context. The headline is the mean over questions. Grounded recall r_i reuses the same matcher on the model’s answer; gain over copy is $r_i - c_i$.

Reading the number. A ceiling near 1 (0.964 here) identifies a *serving regime*: the context already contains the answer strings, so retrieval supplies most coverage and the target is faithful delivery: generative machinery must be measured as gain *above* the ceiling, not against a bare model. A ceiling near 0.5 or below is *consistent with* a synthesis regime, in which answers must be composed across pieces the context does not

name outright and GraphRAG-style traversal, community summaries or multi-hop reasoning can earn their keep; but a low ceiling can equally indicate retrieval failure, vocabulary mismatch (§6 is a direct example), incomplete gold or matcher failure, so it flags that the serving regime has been left, not that the supplied evidence suffices for synthesis. Computing the ceiling first tells you which regime you are in before you build for the wrong one; diagnosing *why* a ceiling is low needs the retrieval-side checks of §6. A worked question, its gold and a paraphrase, and a sample scaffold block are in Appendix B.

6 Vocabulary Mismatch: the Retrieval Boundary

Our serving node retrieves lexically, and a standing limitation was that its red-gated semantic fallback had never been tested on the workload it exists for. We tested it with a paraphrase stress-set: all 510 sweep questions rewritten by an LLM to preserve meaning while excluding the seed class’s title tokens, leak-checked mechanically (0 of 510 contain a title token) and graded for resolvability by two model families (69% resolvable by at least one; results are reported on the full set and that subgroup). The instrument is the copy ceiling itself, computed per arm with no generation: does the retrieved block expose the gold?

Three results. First, vocabulary mismatch collapses lexical exposure: mean ceiling falls from 0.96 on the original questions to 0.34 on the paraphrases, with 62% of questions below 0.5. Second, the failure is *silent*: because partial word overlaps almost always engage some seed, the designed absence-keyed semantic fallback would fire on only 3 of 510 questions: a gate keyed to finding nothing cannot detect finding the wrong thing, and the fallback as designed is therefore inert against the failure mode that actually occurs. (The production façade ships this path unconfigured; the arms here are a harness-side reconstruction using its embedding model and corpus.) Third, the signal the gate wastes is real: a semantic-only arm (top-5 cosine seeds over all 8,146 class embeddings) recovers +0.42 ceiling [+0.38, +0.47] precisely on the failing 62%, while a naive union arm recovers only a third of that: the fixed token budget clamps seeds lexical-first, evicting the semantic seeds that carried the rescue. The engineering prescription is specific: a useful hybrid gate must be keyed to retrieval *confidence*, not presence, and must *replace* weak lexical seeds rather than append to them. Within the paper’s framing, the ceiling again does the work: it locates the boundary of the serving regime (rephrased queries fall out of it), measures what a semantic path could recover, and prices the design that would recover it. A generation-side check on the local model confirms this end to end. On a stratified 100-question generation pass, delivery tracks the exposure ordering: semantic-only answer recall beats lexical exactly on the crossover where lexical fails (lexical ceiling < 0.5, 62% of the set), +0.221

[+0.118, +0.323] with the bootstrap CI excluding zero (recall 0.10 → 0.32), while the naive-union arm’s gain is not significant (+0.030, CI includes zero), reproducing the budget crowd-out in delivered recall. One caveat feeds the gate design: semantic-only’s exposure-to-delivery gap is the widest of the three arms (recall minus ceiling −0.145 on the crossover), so it exposes more gold than the model reliably selects among the extra candidate seeds and its delivered advantage is smaller than its ceiling advantage, a further input for the confidence-keyed gate rather than a retrieval-recall shortfall.

Caveats on the prescription. Three qualifications bound the prescription. First, “replace weak lexical seeds” must be *query-type-aware*. On entity and rare-token queries dense retrieval is the fragile arm and lexical matching the one that holds up [52]: exactly the identifier-, code- and entity-shaped queries a semantic-only replacement would regress. The safe form gates on query type, or on dense-versus-lexical rank disagreement, and fuses the two rankings with reciprocal-rank fusion [6] rather than discarding lexical seeds wholesale. Second, the gate cannot be keyed on intrinsic query-performance prediction: score-distribution predictors are weakest precisely on semantic queries and cannot see a confident-but-wrong lexical seed [13], which is why disagreement between the two retrievers, not a single-ranking confidence score, is the informative key. Third, the collapse reported here is a retrieval-*exposure* measurement (whether the block reaches the model at all), not the model’s answer robustness: the paraphrase-robustness literature reports only single-digit accuracy drops [36] because it measures a different construct downstream of successful retrieval. The 0.96-to-0.34 ceiling fall is not comparable to those figures and should not be read as one.

7 Method

Design. Both studies use within-model pairs: the *same* model answers the *same* question twice, cancelling between-question difficulty variance and yielding a paired per-question estimand. The ten-model sweep varies the *model* with grounding fixed; the new study varies the *serving path* with the model fixed.

Study 1: the ten-model sweep. The knowledge graph supplies templated questions for every sufficiently-defined class: **T-REL** (requirements, uses, dependencies, enablers, and parts), **T-TAX** (immediate and ancestral parents), and **T-COMMON** (shared ancestors of a pair). Gold answers are the {slug, title} targets asserted by the graph. Seed 42 yields **510 questions** across 15 domains: 285 T-REL, 180 T-TAX, and 45 T-COMMON. Each is posed *raw* and *scaffold* (with a budget-clamped 1,500-token ontology extract prepended) to ten models from eight developer families (Google, Anthropic, OpenAI, Zhipu, DeepSeek, Meta, Alibaba, Mistral), accessed through

their providers’ APIs or a router, under one fixed configuration (`temperature=0`, `max_tokens=2048`; `reasoning_effort=low` only where thinking cannot be disabled, preventing mandatory passes from truncating answers). The lexical matcher from §3 counts a gold item when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. Because this undercounts paraphrase, absolute recall is a lower bound; the signal is the explicitly lexical *paired* delta for the same question, scorer, and model. We detected truncation per row: GLM-4.6 truncated on 174 rows, of which 5 were retried and the remainder scored as received, making that row lower-confidence; Gemini 2.5 Flash-Lite truncated on 16.

Study 2: the production-node paired study. We hold the model constant at Qwen3.8-27B on the reference GPU host, isolating the serving path. The *loom* arm calls the production Rust node’s `/v1/chat/completions` endpoint, which retrieves, confidence-gates and injects the scaffold, returning telemetry (engagement, injected tokens, fusion path) in the response *loom* block. The *raw* arm calls the same backend without a scaffold. Both use `temperature=0` and `max_tokens=1536`, below which the reasoning backend truncates output to empty. Three pre-generation frozen sets contain **117** questions: 24 *arcane* (deep in-domain), 33 *thin* (sparsely covered), and 60 *general* (out-of-domain). The general set is also graded by the out-of-domain judged arm below, keeping the judged arms comparable. The four negative-control arms from §3.4 use the same node. Their scaffolds come from the LLM-free `/loom/scaffold` endpoint, ensuring that each control presents the bare model, via a system message, with exactly the block served to the *loom* arm.

Out-of-domain judged arm. A five-model judged arm poses the 60-question *general* set (adjacent, in-domain-general and off-domain subsets) to *harnessed* and *bare* configurations. Each pair is graded 0–5 against frontier-authored key points, providing an independent out-of-domain instrument (§8.6).

Cross-family semantic judge. To cover paraphrase across Study 2’s in-domain and out-of-domain questions, we use a reference-guided **0–5 rubric** judge rather than lexical scoring. The judge is `openai/gpt-4.1` via `OpenRouter` with `temperature=0` and `max_tokens=200`, applying the same fixed system-message rubric in every arm. It is from a different family than the Qwen candidate, limiting self-preference [45, 63]; it is blind to each answer’s arm and study origin, and grading can resume per (set, id, arm). References are graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. Per set and pooled, the primary metric is the paired mean difference with a seeded **10,000-**

resample bootstrap 95% percentile interval [11, 27]; the bootstrap targets the mean difference, and the rank test below targets a different summary, so the two are complementary rather than interchangeable. We report a two-sided **exact conditional signed-rank** test: zero differences are dropped (the reduced-sample procedure, not Pratt’s), tied absolute differences receive average ranks, and the p -value is computed by exact enumeration of the sign-randomisation distribution conditional on the observed ranks. This exact form matters here because one set has only two non-zero pairs, where the usual normal approximation is invalid. We also report the paired effect size, **matched-pairs rank-biserial correlation** $r = (T^+ - T^-)/(T^+ + T^-)$, over non-tied pairs; this is preferable to Cliff’s delta for matched pairs because Cliff’s independent-samples dominance statistic divides the win–loss margin by the many ties. Set-level p -values receive a **Holm–Bonferroni** correction [8]. For out-of-domain non-regression we use a conservative *interval-inclusion* criterion in the spirit of equivalence testing [29]: non-regression is declared only when the 95% bootstrap percentile interval lies wholly inside a pre-specified ± 0.25 margin (conventional TOST at $\alpha = 0.05$ corresponds to a 90% interval, so requiring the 95% interval is stricter); we do not report one-sided TOST p -values. Because in-domain graph-derived questions cluster by class and domain [8], we report a domain-clustered block bootstrap alongside the naive interval and an intention-to-treat delta retaining questions where the gate did not engage [23].

8 Results

Reporting deviations (Study 2). Two deviations affect interpretation of the production-node figures. (i) *Budget-exhaustion re-runs.* At `max_tokens=1536`, 42 of 234 *loom/raw* completions (117 questions $\times$ two arms), spanning 28 distinct question pairs, exhausted the budget during reasoning and produced no output. Each affected pair was re-executed at 4096 in *both* arms; three questions (all *thin*) still produced an empty answer in one arm — *loom* once, *raw* twice, so realised missingness was not symmetric — and were excluded pairwise, yielding a complete-case $n = 114$ paired analysis (*thin*: 33 to 30). The pooled contrast is therefore a mixed-budget estimand: 25 surviving re-run pairs use 4096 and the remaining 89 use 1536 (counts verified against the released per-row data). (ii) *Controls: originally attrition-biased, since rerun.* The four negative-control arms originally remained at 1536 and had substantial empty attrition: irrelevant 28/56 (50.0%), true 24/56 (42.9%), masked 23/56 (41.1%), and shuffled 20/56 (35.7%). The rerun of §8.5, under a common retry policy (4096 tokens, stragglers retried at 8192), eliminates this (0% attrition) and, re-judged under a single model with a length-matched fluent-noise arm, revises the earlier content-specificity reading; §8.5 carries the corrected result.

8.1 In-domain: delivery fidelity

For Gemini 3.7 Flash, raw recall is 0.375 and scaffold recall 0.942, a paired delta of **+0.567** with naive and domain-clustered 95% CIs of $[+0.532, +0.603]$ and $[+0.524, +0.611]$, respectively, across $n = 510$ items; there were no errors or truncated responses (Figure 2). The low raw score is consistent with a domain only partially covered by pretraining (the corpus is public, so some exposure is expected; §1). The copy ceiling is **0.964**, placing recall -0.022 below the fraction of gold already exposed. Thus the model returns most curated answers but omits some; because recall does not measure additions beyond the gold, fabrication and attribution precision remain unmeasured (§17.1). §8.2 tests whether models recover lexically unexposed gold or drop exposed facts.

8.2 Ten models: gain over copy

Table 1 applies the paired design to ten models from eight developer families under one fixed configuration. Raw→scaffold uplift ranges from $+0.57$ to $+0.78$, while scaffold recall spans 0.897–0.942 against the model-independent ceiling of **0.964**. Hence raw uplift does not distinguish delivery from exposure.

Signed **gain over copy** is negative throughout (Figure 3), from -0.067 for Gemini 2.5 Flash-Lite to -0.022 for Gemini 3.7 Flash. Because $g_m = R_m - 0.964$, ranking by gain equals ranking by scaffold recall; its contribution is an interpretable scale, not a new ordering. Recent models sit nearest the ceiling, while Gemini 2.5 Flash-Lite, GPT-4.1-mini and Mistral-Small-24B fall further below, an ordering specific to this task. Two models had transport-level truncations under the fixed 2048-token budget: GLM-4.6 had 174 truncated and 5 retried, with the rest scored as received and its row treated as lower-confidence; Gemini 2.5 Flash-Lite had 16 truncated.

What adds information: the exposure/recovery decomposition. A negative g alone cannot distinguish omission of exposed gold from failure to recover unexposed gold. We therefore classify each item by scaffold exposure and answer recovery. Across ten models and 11,360 gold items, $n_{11} = 9,865$ were exposed and recovered, $n_{10} = 735$ exposed and omitted, $n_{01} = 3$ unexposed and recovered, and $n_{00} = 757$ unexposed and omitted (Table 2). Thus only **3 of 11,360** items (0.026%) were recovered without exposure, with no model exceeding $n_{01} = 1$. With $n_{01} \approx 0$, $g \approx -n_{10}/|G|$: the deficit chiefly measures imperfect copying, including lexical-match under-counting of paraphrase, rather than reasoning. Context utilisation, $n_{11}/(n_{11} + n_{10})$, ranges from 0.900 to 0.955, implying roughly one exposed item in fourteen is dropped. It is micro-averaged over exposed items, whereas recall averages per-question ratios with varying c_i ; accordingly, the rankings differ once: llama-3.3-70b has utilisation 0.934 versus deepseek-chat’s 0.931, but recall 0.916 versus 0.924.

8.3 Production-node paired study

Study 2 holds Qwen3.8-27B and decoding constant on the reference host while varying only whether requests pass through the shipped Loom node (*loom*) or reach the backend directly (*raw*). The node includes live retrieval, confidence gating and injection authority; a cross-family GPT judge scores answers 0–5 as described in §7. Table 3 reports paired results.

Across $n = 114$, loom raises judged quality by **+0.27** points ($[+0.11, +0.45]$, exact signed-rank $p = 0.0023$, matched-pairs rank-biserial $r = +0.589$), with 24 wins, 8 losses and 82 ties, so the pooled effect rests on the 32 non-tied pairs, the other 82 scoring identically in both arms. The arcane-set effect is **+0.79** ($[+0.17, +1.42]$, exact $p = 0.032$, Holm-adjusted $p = 0.097$, $r = +0.670$); the thin-set estimate is $+0.30$ ($[-0.03, +0.63]$), positive but non-significant. For general questions, $+0.05$ $[0.00, +0.13]$ lies within the pre-specified ± 0.25 margin, a non-regression reading by interval inclusion (§8.6); its exact p is 0.5 on only two non-zero pairs, and with 58 of 60 pairs tied (mean scores 5.0 loom, 4.95 raw) the instrument is near its ceiling and insensitive on this set. The three set-level point estimates ($+0.79$, $+0.30$, $+0.05$) fall in curation-depth order, but their intervals overlap, so the gradient is three point estimates with overlapping intervals, suggestive rather than established; no interaction test separates a curation-depth effect from other differences between the sets (Figure 4).

Injection engagement is 95.8% on arcane, 100% on thin and 78.3% on general. Thus the general-set null cannot be explained chiefly by the gate skipping questions, although it skipped the remaining 21.7%. Median injected context is $\approx 1,200$ – $1,300$ tokens across sets. Median latency is comparable for arcane, 89.8 s (loom) versus 93.4 s (raw), and general, 66.0 versus 65.7 s. Thin is slower under loom, 172.4 versus 145.2 s, consistent with full engagement and longer scaffolds.

8.4 Judge-family robustness

Study 2’s loom-versus-raw contrast is graded by a single **gpt-4.1** judge, one provider family. Because the judged studies elsewhere in this paper draw on OpenAI and Google judges (Table 6), a residual concern is that the pooled uplift tracks a family-specific grading style rather than answer content. To test this, the same 114 paired answers were re-judged from scratch by **claude-opus-4-6**, a judge family used nowhere else in the paper, under the identical rubric, blinding and resume protocol. The contrast reproduces and slightly strengthens under the new family: pooled $+0.342$ ($n = 114$, 33 non-tied) against the **gpt-4.1** $+0.272$, with the same per-set ordering (Table 4). The arcane effect rises to $+1.042$ (from $+0.792$) and the thin effect to $+0.400$ (from $+0.300$), while the general set stays near zero ($+0.033$ versus $+0.050$). The general-set estimate continues to rest on two to three non-tied pairs, so the thin general-set caveat of §8.6 is unchanged. The direct robustness check is therefore

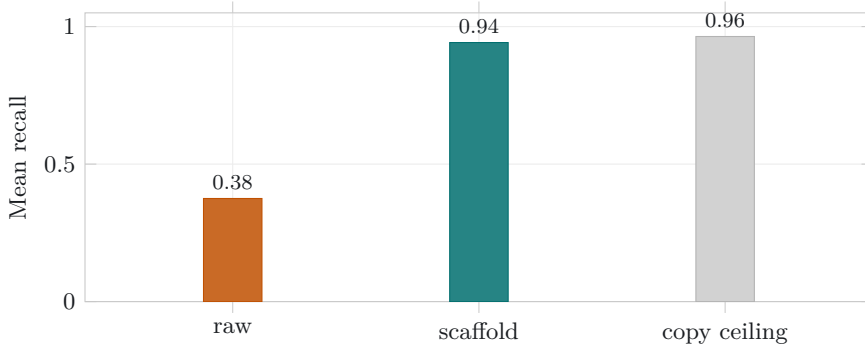


Figure 2: Gemini 3.7 Flash, $n = 510$. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964), the recall a no-op extractor of the injected context would already achieve. The $+0.57$ raw $\rightarrow$ scaffold uplift is almost entirely answer exposure, not reasoning over structure.

Table 1: Ten models, one fixed configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall – copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

OpenAI versus Anthropic on the same 114 answers, which agree on sign and approximate magnitude; the Google-judged control rerun of §8.5 grades a different cohort and contrast and is consistent in sign but not directly comparable. Different provider families limit shared grading style; they do not establish statistical independence.

8.5 Negative controls: attributing the gain (uniform-budget rerun)

Four controls test whether the live lift reflects specific knowledge transfer or merely the presence of a fluent, on-corpus block. Each holds model and question fixed while perturbing only the scaffold fetched verbatim from the LLM-free /loom/scaffold endpoint (§3.4). The original arms suffered 36–50% empty-answer attrition at the 1536-token budget, leaving small, unequal survivor sets. We reran all four — plus a fifth, **fluent-noise** (a length-matched block of random on-corpus sections, the length-matched-noise confound raised by Cuconasu, Trappolini, Siciliano, et al. [7]) — under a common retry policy: 4096 tokens, with the few stragglers retried at 8192 (per-question final budgets therefore vary within arms). Attrition fell to **0% across all five arms**; the cause was confirmed as reasoning-budget exhaustion, the backend emitting a reasoning block before its answer so that at 1536 tokens the reasoning alone consumed the budget. To separate

the budget/attrition fix from judge variance, both the original complete-case rows and the new intention-to-treat (ITT) rows were re-judged from scratch under a single judge, **gemini-3.1-pro-preview**, using the paper’s rubric verbatim, with Holm–Bonferroni correction across the eleven-contrast family; the original **gpt-4.1** figures are retained as historical context (Figure 5). Study 2’s $+0.27$ headline (§8.3) remains under its original **gpt-4.1** judge and was *not* re-judged by this Gemini control judge (its separate third-family re-judge is §8.4) — a different instrument, whose scale is never mixed with these Gemini-judged control figures in any comparison. One envelope difference also bears stating: the live node injects the block into the last user message, whereas the control arms supply the same bytes as a system message, so identical block content does not by itself isolate every serving-path effect. Table 5 reports the dual-judged comparison.

What survives, and what does not. One contrast holds in both readings: **true – raw** = $+0.75$ [$+0.36, +1.17$] (ITT), Holm-significant: the ontology scaffold as a whole reliably beats no context. The content-specificity contrasts do not survive once the fluent-noise floor is in place and attrition is fixed. **True – irrelevant** is $+0.28$ [$−0.07, +0.65$] and crosses zero even at $n = 54$, double the original sample; **true – fluent_noise** is $+0.25$ [$−0.02, +0.55$]; and **fluent_noise – irrelevant** is a null $+0.02$ [$−0.24, +0.30$].

Table 2: Exposure/recovery decomposition over the 510-question sweep (1,136 gold items per model; 11,360 pooled), sorted by context utilisation. Utilisation is $n_{11}/(n_{11} + n_{10})$, the fraction of *exposed* gold the answer recovers; n_{10} is exposed gold the answer omitted; n_{01} is gold recovered though the scaffold did not expose it. Across all ten models n_{01} never exceeds 1, and is 3 pooled: recovery beyond exposure is empirically negligible. Counts are pooled over gold items (micro-averaged), whereas the headline recall and ceiling average per-question ratios (macro-averaged), so these counts do not reconstruct those numbers directly.

Model	Context utilisation	n_{10} (exposed, omitted)	n_{01} (unexposed, recovered)
Gemini 3.7 Flash	0.955	48	0
Gemini 3.5 Flash-Lite	0.947	56	0
Claude Haiku 4.5	0.946	57	1
GLM-4.6	0.945	58	0
Llama-3.3-70B	0.934	70	0
DeepSeek-Chat	0.931	73	0
Qwen-2.5-72B	0.924	81	1
Mistral-Small-24B	0.917	88	1
GPT-4.1-mini	0.908	98	0
Gemini 2.5 Flash-Lite	0.900	106	0
Pooled	0.931	735	3

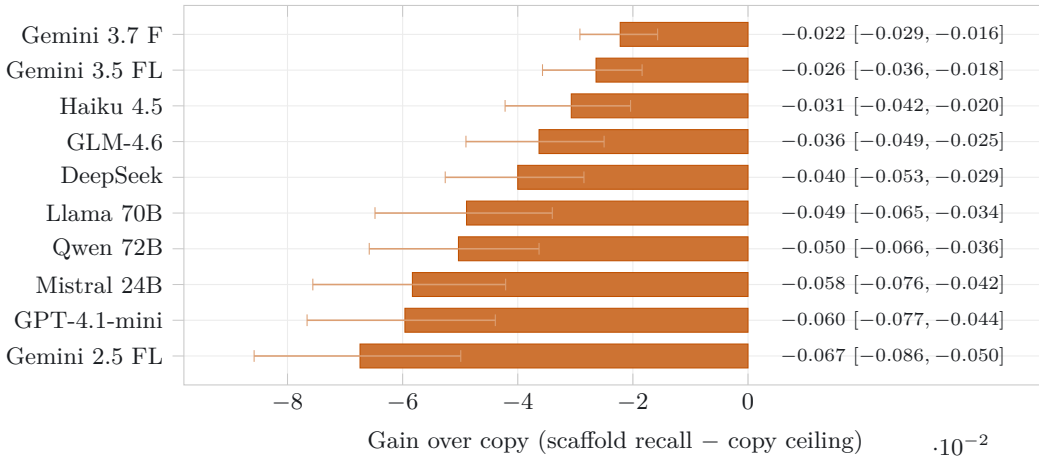


Figure 3: Signed gain over copy for all ten models, negative throughout, with per-model seeded 10,000-resample bootstrap 95% intervals (whiskers); the value column at right gives each point and interval, rounded independently from the full-precision artifact (points, whiskers and labels are all generated from it). No model’s interval crosses zero, and none exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) puts the models on an exposure-normalised scale that raw uplift (+0.57 to +0.78) does not provide. Models nearest zero (top) restate the vetted facts most faithfully.

Structure fares no better: **true** − **shuffled** *flips negative* in the ITT data (−0.23 [−0.45, −0.02]), and **shuffled** − **raw** = +0.94 is the single strongest uplift in the table, the only contrast besides true − raw to clear Holm. Sentence order thus appears to carry none of the effect on this instrument: destroying it does not lower the uplift and, if anything, raises it. Fixing the attrition did not rescue content-specificity: with attrition eliminated the content-specificity contrasts remain inconclusive, so the earlier complete-case impression is unsupported — though these data show non-establishment, not absence, of a content effect. The supported claim is therefore narrower than our earlier reading: *ontology-shaped context helps over no context, but whether the specific retrieved content matters beyond generic on-corpus text is not established by these controls*. This is the length-matched-noise confound Cuconasu, Trappolini, Siciliano, et al. [7] warn of; our fluent-noise arm is designed to expose it,

and the observed nulls are consistent with it.

Placebo-verification lesson stands as method.

An initial shift-by-one derangement kept donor and target within the same domain cluster, preserving neighbouring vocabulary; enforcing seed-IRI disjointness, each donor sharing neither a seed IRI nor seed class with the target gold, is what removed that leakage and dropped the irrelevant arm’s estimate toward the fluent-noise floor. That procedure remains a valid control-design lesson independent of the downgraded effect: an unverified placebo would have carried donor vocabulary and produced a misleadingly favourable baseline. We no longer read the verified-irrelevant arm as evidence of content-specific transfer, only as a correctly specified floor.

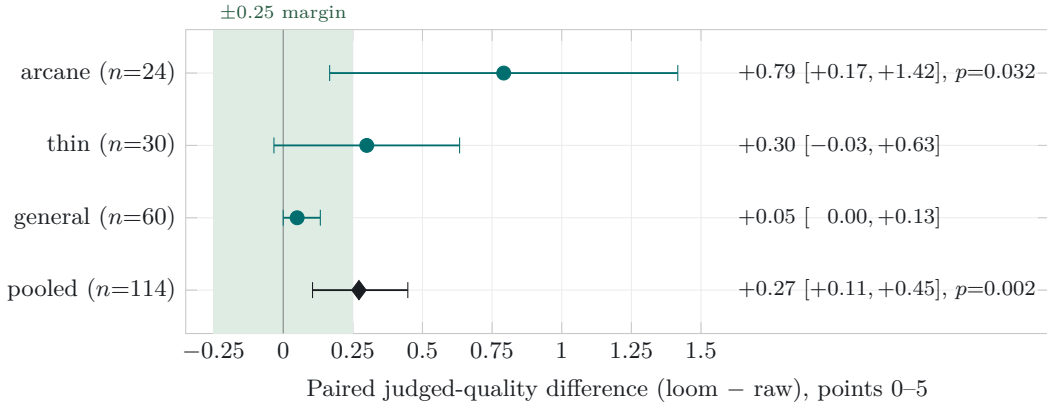


Figure 4: Study 2 paired judged-quality deltas (Qwen3.8-27B, model held constant, loom versus raw serving path; cross-family GPT judge), per set and pooled, with seeded 10,000-resample bootstrap 95% intervals. The three sets are ordered by curation depth: the effect is largest on the deep *arcane* set (+0.79), attenuates on the sparsely-covered *thin* set (+0.30), and is negligible out-of-domain on *general* (+0.05). The three point estimates fall in curation-depth order, but their intervals overlap and no interaction test separates a curation-depth effect from other differences between the sets, so the gradient is suggestive rather than established. The shaded band is the pre-specified ± 0.25 margin: the general-set interval $[0.00, +0.13]$ sits wholly inside it, so out-of-domain non-regression is read by interval inclusion, not from a non-significant difference; with 58 of 60 general pairs tied the instrument is insensitive on that set (§8.3, §8.6).

Table 3: Study 2, the production-node paired study: Qwen3.8-27B, model held constant, loom versus raw serving path, judged 0–5 by a cross-family GPT judge. Paired mean difference with seeded 10,000-resample bootstrap 95% CI, two-sided exact conditional signed-rank p (Holm-adjusted across the three sets), the matched-pairs rank-biserial r (paired effect size, conditioned on non-tied pairs), and win/loss/tie counts. Complete-case after budget-exhaustion re-runs (§8).

Set	n	Δ (loom–raw)	95% CI	p (Holm)	r	W/L/T
arcane	24	+0.79	[+0.17, +1.42]	0.032 (0.097)	+0.670	10/3/11
thin	30	+0.30	[−0.03, +0.63]	0.130 (0.261)	+0.431	12/5/13
general	60	+0.05	[0.00, +0.13]	0.500 (0.500)	+1.000 [†]	2/0/58
pooled	114	+0.27	[+0.11, +0.45]	0.0023	+0.589	24/8/82

[†] The general-set r rests on 2 non-tied pairs (58 ties); read it as directional only, not as a magnitude.

Table 4: Judge-family robustness for Study 2: paired mean difference (loom–raw), 0–5 scale, graded independently by *gpt-4.1* (OpenAI, the original judge) and *claude-opus-4-6* (Anthropic, a family used nowhere else in the paper). Same 114 pairs, same rubric. The contrast reproduces across families with matched per-set ordering.

Set	gpt-4.1 Δ	opus-4.6 Δ
arcane	+0.792	+1.042
thin	+0.300	+0.400
general	+0.050	+0.033
pooled	+0.272	+0.342

the interval-inclusion result: general-set $\Delta = +0.05$ $[0.00, +0.13]$ lies within the pre-specified ± 0.25 margin. That single-model result does not generalise to the five-model arm: the worst-case subgroup interval $[-0.63, +0.05]$ extends beyond the margin, so equivalence is *not* established there, only not-detected regression. Two further cautions: the general-set instrument is blunt here (58/60 Study-2 pairs tie, with mean scores 5.0 and 4.95, so a real but small effect could hide in the ties), and non-detection under a gate that engaged on 78.3% of general questions is evidence about the deployed policy, not proof that the gate causes the non-regression — there is no matched always-inject ablation.

8.6 Out-of-domain: the gate holds

In the five-model judged arm (§7), a cross-family judge scored 60 out-of-domain questions 0–5 against frontier gold. Every harness–bare 95% interval includes zero: all questions -0.05 $[-0.11, 0.00]$, off-domain -0.09 $[-0.25, +0.02]$, and the pre-specified worst case of erroneous injection -0.23 $[-0.63, +0.05]$ (Table 7). The gate correctly skipped 60 of 100 off-domain gradings, making harnessed and bare answers identical; detected degradation was confined to over-firing and concentrated on the weakest model. Study 2 provides

9 Analysis: Exposure, not Reasoning

Both studies indicate that graph-grounding uplift over a curated corpus is almost entirely *exposure*: faithful delivery of curated facts shown to the model, rather than reasoning over injected structure. The instrument establishes this distinction and yields two engineering implications. One scope condition holds throughout:

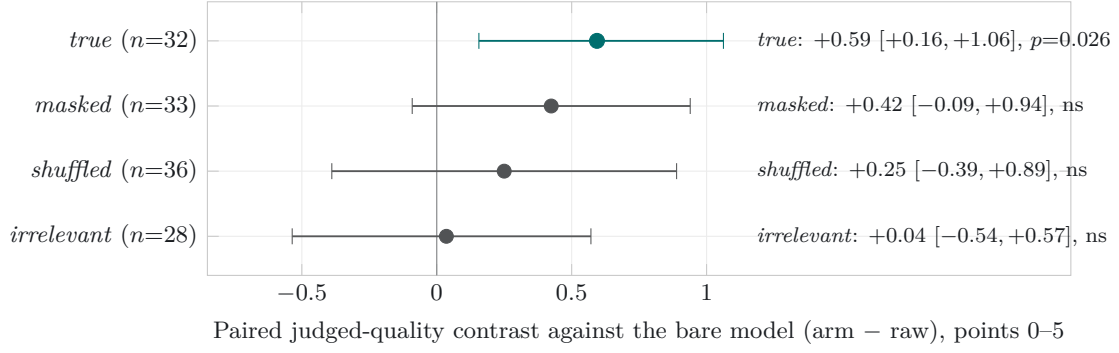


Figure 5: Negative-control decomposition, *historical*: the original complete-case contrasts (Study 2, pooled arcane + thin, 1536-token budget, **gpt-4.1** judge), each arm holding the model and question fixed and perturbing only the injected block, read as a paired contrast against the bare model (arm – raw) with seeded bootstrap 95% intervals (*true* – raw +0.59; *irrelevant* – raw +0.04; *true* – *irrelevant* +0.35). This complete-case snapshot suffered 36–50% empty-answer attrition and read as content-specific; it is retained for context only. The uniform-budget, single-judge rerun with a length-matched fluent-noise arm (Table 5) supersedes it: with attrition removed and Holm correction applied, only *true*–raw and shuffled–raw survive and the content-specificity contrasts do not (§8.5).

Table 5: Negative-control contrasts, pooled arcane + thin, judged 0–5, paired mean difference with seeded bootstrap 95% CI. Both columns are judged by the same model (**gemini-3.1-pro-preview**, paper rubric verbatim) so the comparison isolates the budget/attrition fix: *old* is the original complete-case rows (1536-token budget, varying *n*); *new* is the uniform-budget intention-to-treat rerun (4096-token budget with 8192-token retries, 0% attrition). [†] marks contrasts surviving Holm–Bonferroni across the eleven-contrast family. Only *true*–raw (both readings) and shuffled–raw (ITT) survive; no content-specificity contrast does. Each contrast is computed on its own pairwise-complete question set, so a direct contrast need not equal the difference of the two arm–raw rows; per-contrast *n* varies with arm pairing.

Contrast	Old (complete-case) Δ [95% CI]	New (ITT) Δ [95% CI]
true – raw	+0.78 [+0.31, +1.28] [†]	+0.75 [+0.36, +1.17] [†]
shuffled – raw	+0.22 [–0.31, +0.75]	+0.94 [+0.56, +1.33] [†]
masked – raw	+0.47 [+0.06, +0.94]	+0.62 [+0.20, +1.04]
irrelevant – raw	+0.21 [–0.29, +0.71]	+0.44 [+0.10, +0.79]
fluent_noise – raw	—	+0.46 [+0.17, +0.78]
true – irrelevant	+0.30 [–0.22, +0.91]	+0.28 [–0.07, +0.65]
true – shuffled	+0.43 [+0.13, +0.77]	–0.23 [–0.45, –0.02]
true – masked	+0.27 [–0.08, +0.65]	+0.08 [–0.20, +0.35]
true – fluent_noise	—	+0.25 [–0.02, +0.55]
fluent_noise – irrelevant	—	+0.02 [–0.24, +0.30]
loom – true	–0.06 [–0.44, +0.31]	–0.22 [–0.54, +0.09]

the ≈ 0.92 grounded recall is for questions phrased in the graph’s title vocabulary; under paraphrase the exposure ceiling itself falls to 0.34 (§6), and real user queries sit between these extremes, so the headline delivery figure is the high end of a distribution, not a single operating point.

9.1 Grounding suppresses parametric recall

This is the one result in the paper that needs no ceiling to state: it is a bare-versus-grounded comparison on the $n_{01} + n_{00}$ items the scaffold does not expose, and it is the finding with the most direct operational consequence. A negative gain over copy admits two readings: either the model adds nothing, or the ceiling simply out-runs it. Restricting attention to exactly the gold items the scaffold does *not* expose sharpens this into a positive finding. Bare, those items are not beyond the models: pooled over the ten models and

the 760 unexposed gold items (76 per model), the raw arm recovers **12.1%** of them from parameters alone (92 of 760). Under injection the *same* items are recovered at **0.4%** (3 of 760), a roughly **30×** reduction, and seven of the ten models drop to *exactly zero*. Recovery without injected context therefore exceeds recovery with it on these items: the observed behaviour is suppression. Displacement of parametric knowledge by injected context, a documented tendency [35, 54, 62], is the natural explanation, though absence of an item from the scaffold is omission rather than contradiction, and this design observes the effect rather than isolating its mechanism. Three caveats bound the reading: the shared lexical matcher makes both rates lower bounds on paraphrased recovery; a bare-arm hit does not prove purely parametric provenance (lexical cues in the question can contribute); and the effect is a property of this confidence-gated injection policy rather than of grounding in the abstract. The figures are computed with the paper’s own byte-identical matcher over the

Table 6: Judge provenance across the judged studies, for auditing family overlap. Study 1 uses no judge (deterministic lexical scoring). A † marks a judge sharing a provider family with a candidate model it graded. The five-model out-of-domain arm (§8.6) uses the cross-family judge described in §7.

Study	Judge (family)	Candidate families judged	Reference source
Study 1 (ten-model sweep)	none — lexical, deterministic	Google, Anthropic, OpenAI, Meta, DeepSeek, Alibaba, Mistral, Zhipu	graph-derived {slug,title}
Study 2 (paired serving path)	gpt-4.1 (OpenAI)	Qwen/Alibaba (single candidate; no family overlap)	graph-derived in-domain; frontier-authored out-of-domain
Study 2 re-judge (§8.4)	claude-opus-4-6 (Anthropic)	Qwen/Alibaba	same 114 pairs, same rubric
§8.5 controls (re-judged)	gemini-3.1-pro-preview (Google)	Qwen/Alibaba	graph-derived
Write-path page quality (§10.1)	gemini-3.1-pro-preview (Google) primary; gpt-5.6 (OpenAI) 20% secondary	ten serving arms across six model families incl. Google†, OpenAI†	the pages themselves (blind before/after)

Table 7: Out-of-domain non-regression (five-model judged arm): paired harness—bare quality delta (judge 0–5 vs frontier gold), five models, seeded paired bootstrap 95% CI. Every interval includes zero. The worst case is off-domain questions where the confidence gate mis-fired and injected anyway; the last row is where it correctly skipped.

Subset	Mean Δ	95% CI	n pairs
All general questions	−0.05	[−0.11, 0.00]	300
adjacent	+0.01	[−0.02, 0.05]	100
in-domain-general	−0.06	[−0.14, 0.00]	100
off-domain	−0.09	[−0.25, 0.02]	100
Off-domain, gate mis-fired (worst)	−0.23	[−0.63, 0.05]	40
Off-domain, gate skipped (correct)	+0.00	n/a	60

released sweep rows (per-model raw recovery 1–14 of 76). This is the more informative of the two observations: not merely that grounded recall fails to exceed a verbatim copy (negative gain), but that supplying the context is followed by loss of recall the bare model demonstrably has.

9.2 Delivery fidelity and its consequences

Delivery fidelity is the headline, and the instrument is what reveals it. In the ten-model sweep, raw uplift is large and consistent (+0.57 to +0.78), whereas gain over copy is small and uniformly negative (−0.067 to −0.022). Every model appears transformed against its bare-model baseline, but none exceeds a verbatim copy of its context; they differ only in how much exposed gold they drop. Uplift therefore reveals little about model performance, while gain over copy places the same ordering on an interpretable, exposure-normalised scale (with a constant ceiling it preserves the recall ranking; its contribution is the reference point, not a new ordering). This supports reporting the ceiling as standard practice.

Live controls attribute less than first appeared. A uniform-budget rerun with a length-matched fluent-noise floor and a single judge revises our earlier read-

ing (§8.5). The scaffold as a whole reliably beats no context (true—raw +0.75, Holm-significant), but no content-specificity contrast survives correction: true—irrelevant (+0.28), true—fluent-noise (+0.25) and fluent-noise—irrelevant (+0.02) all cross zero, and true—shuffled flips negative. The graded facts therefore cannot be shown to carry the effect beyond what any fluent, on-corpus block of the same length supplies: the length-matched-noise confound of Cuconasu, Trappolini, Siciliano, et al. [7], met here directly rather than assumed away. What remains solid is the coarser attribution: injecting ontology-shaped context helps, while the earlier complete-case impression of *content*-specificity does not survive the corrected design; whether a content effect exists below the sensitivity of these controls is left open. The seed-IRI-disjointness step still matters as method (an unverified placebo leaked donor vocabulary and inflated the baseline), but we now read the verified-irrelevant arm as a correctly specified floor, not as proof of content transfer.

Consequence 1: the copy ceiling is a regime detector. Compute it before you architect. Calculate the ceiling on representative questions before selecting an architecture. A ceiling close to 1 (0.964 here) indicates that answer entity names are exposed in the context, although it does not prove that the required relation is present or understood; the follow-

ing regimes are therefore heuristic. A high ceiling identifies a *serving regime*: retrieval supplies most target coverage, while generative re-encoding may lose fidelity, though generation can still improve selection, relevance and presentation. The target is faithful delivery, measured by gain over copy. A low ceiling identifies a *synthesis regime*, in which answers must be composed across retrieved pieces and GraphRAG-style machinery (community summaries, multi-hop traversal, GNN soft-prompts) may add value. That value must be measured above the ceiling rather than against a bare-model baseline. In a high-ceiling setting, bare-model uplift is delivery masquerading as reasoning. We suspect much reported graph-RAG gain reflects undiagnosed serving-regime exposure; the ceiling distinguishes the regimes before expensive architecture is built.

Consequence 2: value concentrates at curation, and the instrument doubles as a promotion gate. When the model delivers answers supplied by the corpus, investment should favour upstream curation, whose cost is amortised rather than repaid per query. Automated extraction candidates, such as blocks proposed by OntoCast [19] (analysed here solely as a third-party upstream producer), can be evaluated through the copy ceiling of the questions they answer. A block that drives the ceiling towards 1 is answer-complete and eligible for promotion; one that does not is unready. Thus the serving-layer audit also grades ingestion, with a block’s answer-completeness approximated by its copy ceiling. This governance gate is necessary but insufficient: term-stuffing answer names can inflate the ceiling while leaving a block incoherent or wrong. Exposure coverage must therefore complement existing CI, consistency and human-review gates.

10 A Production Case Study: Podcast Assertion Extraction

A production deployment tested whether scaffold grounding helps a model *write into* the corpus. The node’s podcast-ingest pipeline extracts tiered, source-attributed assertions from episode transcripts; each carries `ontology_terms`, concept labels used to locate a destination page. Assertions whose terms resolve to no page are discarded or queued for page proposal, making *term resolution against the live graph* the landing metric. We report a lexical proxy: normalised string matches between emitted terms and the inventory of 8,142 classes (this generation snapshot; the read-path studies use the 8,146-class generation of §13, the small difference being ordinary build-to-build churn). This is not a human judgement of assertion quality.

We ran the unmodified production prompt over ten episodes of a technology news podcast (November 2025–August 2026) across ten arms: the node’s local Qwen3.8-27B *through the Loom* (scaffold injected at

the default 1,500-token budget, verbatim short-circuit disabled) and *bare*; and eight frontier or near-frontier cloud models, all bare, because the Loom grounds only the model behind its own façade. All arms used direct chat-completion calls with identical sampling (temperature 0.2, 12,288 max tokens). Claude and GPT used a model router; the others used their providers’ APIs. One Loom call produced three assertions for one episode and remains in the totals.

Holding weights, prompt and GPU fixed, the paired scaffold effect is large: term resolution rises from 0.17 to 0.55 (3.2×) while assertion count falls. Grounding narrows the model’s vocabulary towards the graph’s, as integration requires. Generation time also roughly halves (173s vs. 346s per episode), consistent with the scaffold displacing open-ended reasoning. Grounded, the local 27B exceeds every bare frontier arm (best: 0.42), reproducing Study 2’s read-path pattern in a write path. This does not establish that frontier models are worse extractors: several produce more assertions, and assertion quality is not judged here. Without the graph’s vocabulary, however, their concept labels miss it and the assertions do not land. Pipeline quality also depends on fidelity to noisy input. Several arms reproduce speech-recognition errors as entity names (7/181 for GLM-4.7; 3 each for DeepSeek and Opus 4.8), whereas the Loom-grounded Qwen arm emitted none and the bare Qwen arm one. Corrupted names damage page matching and downstream dedup fingerprints. Figure 6 shows all three observations on one panel.

Our prior production default disabled scaffold injection to work around the façade’s verbatim short-circuit, which answered transcript-shaped prompts from the scaffold without invoking the model. The pipeline now requests injection while explicitly declining that short-circuit. This remains *single-configuration estimation*: one podcast, ten episodes, one run per arm and one lexical scorer yield a point estimate from one draw of one setup, with no sampling variance or transfer test across corpora or scorers. The paired Loom-versus-bare contrast is least exposed because both arms share the draw; another draw could more plausibly reorder the cross-model rows. This is evidence about where extracted knowledge lands, not a model-quality leaderboard.

10.1 Judged page: an end-to-end quality gate

Term resolution does not show whether a page improves. For each arm, we therefore applied its matched assertions to sandbox copies of target pages using the production integration mechanic. A cross-family judge (Gemini 3.1 Pro, temperature 0, blind A/B with seeded order randomisation, un-blinded at analysis) scored each before/after pair. A second judge, GPT-5.6, re-scored a 20% subset with 93% verdict agreement. Every arm degrades judged page quality (means -1.04 to -0.44); the two arms adjacent to a judge’s family are flagged in the released analysis.

Insertion mechanics alone do not explain the result.

Table 8: Assertion extraction across ten serving arms, ten episodes. *Ground* is the fraction of emitted `ontology_terms` that resolve to an existing knowledge-graph page (lexical proxy); *Artif.* counts claims that reproduce speech-recognition artefacts as entity names (e.g. “Opus 48”, “GPT55”). *n* is total assertions; *Conf.* is the model’s own mean stated confidence, which is self-reported and not comparable as a quality measure across families.

Arm	<i>n</i>	Conf.	Ground	Artif.	s/episode
Loom + Qwen3.8-27B (local)	125	0.83	0.55	0	173
Qwen3.8-27B bare (local)	170	0.81	0.17	1	346
Gemini 3 Flash (preview)	107	0.80	0.42	1	13
GLM-4.7	181	0.80	0.39	7	17
DeepSeek v4 Flash	131	0.86	0.28	3	66
Claude Opus 4.8	171	0.75	0.27	3	51
Claude Haiku 4.5	295	0.82	0.22	0	242
GPT-5.6	133	0.82	0.21	0	23
GLM-5.3	169	0.82	0.21	0	33
Claude Sonnet 5	175	0.76	0.20	0	50

A section-splice rewriter emits only an anchored section edit, which deterministic code applies fail-closed while preserving the original by construction. Prompt optimisation covered four variants on a dev split, followed by validation on held-out pages under two rubrics. One rubric was written post-hoc, disclosed as such, to credit informativeness explicitly. The best variant remained negative under both judges (−0.40 dev, −0.90 held-out). Persistence across mechanic, rubric and judge suggests a structural problem: narrow, newly extracted facts dilute a mature curated page.

Under a pre-declared decision rule, the integration phase was disabled while extraction and verification continue, and redesign addresses *placement* rather than phrasing. This is the outcome the promotion-gate discipline of §9 anticipates: the gate identified a value-destroying step before it ran at scale.

What the judged gate can and cannot rule out. Two known judge biases bear on this result, in opposite directions. Contextual judging is itself error-prone: ContextualJudgeBench shows even strong judges struggle on context-grounded verdicts [58], so the degradation is measured with an imperfect instrument. But the most relevant bias would, if anything, mask the finding: judges score text labelled as “refined” more favourably [61], whereas our before/after pairs are blind with seeded order randomisation, so a refinement-aware bias cannot manufacture a *negative* delta. The phenomenon we observe has a longer pedigree in psychology: the dilution effect, whereby non-diagnostic detail weakens a judgement built on diagnostic evidence [41], offers a mechanism consistent with narrow extracted facts eroding a mature curated page, though style, placement, length and truth of the inserted content are not separated here. We do not claim to have excluded every confound: inline enrichment can alter formatting and length, and judges carry style and verbosity preferences that our design controls only partially. A replacement study at constant token budget, separating content from format, is the control

that would close this gap, and it remains future work.

11 Two-Edge Composition: Locating the Synthesis Boundary

Our exposure/recovery decomposition found beyond-exposure recovery in only 3 of 11,360 gold items, but did not test whether models *would* compose fully exposed ontological structure when required. The multi-hop design sketched in §19 does. We mined two-edge chains $A \xrightarrow{r_1} B \xrightarrow{r_2} C$ over the restriction-encoded relations *requires* and *hasPart*. Eligible chains had no direct asserted or inferred $A \rightarrow C$ edge, omitted *C* from *A*’s served block, and had at most three valid targets. Of 6,514 qualifying chains, 150 were sampled across the four relation-pair shapes. Questions name only *A* (“Name a part of something *A* requires”). Every context-bearing arm serves exactly four seed-shuffled blocks (the raw arm serves none), so presence does not signal relevance. The six arms provide *both* premises (*A*’s and *B*’s blocks plus two seed-disjoint distractors), either premise *alone* (plus three distractors), *masked* premises (both with relation labels replaced), *irrelevant* context (four distractors), or *raw* input (no context). Twelve models ran all arms under identical sampling.

Our controls behave as required: *a*-only, irrelevant and raw recall are at floor for every model (zero in almost all cells; none exceeds 2/150), showing no parametric leakage or placebo effect. Because *b*-only exposes the answer title verbatim, the copy metric cannot separate these arms; the informative quantity is the *contrast* $\Delta = \text{both} - \text{b-only}$, which nets out answer exposure and isolates the contribution of route information — though adding *A*’s block also changes relevance cues and retrieval selection, so Δ is a controlled contrast, not a pure readout of internal reasoning. Figure 7 shows three groups, not a capability ladder. Five models compose (positive bootstrap CI): Gemini 3 Flash ($\Delta = +0.33$), the local Qwen3.8-

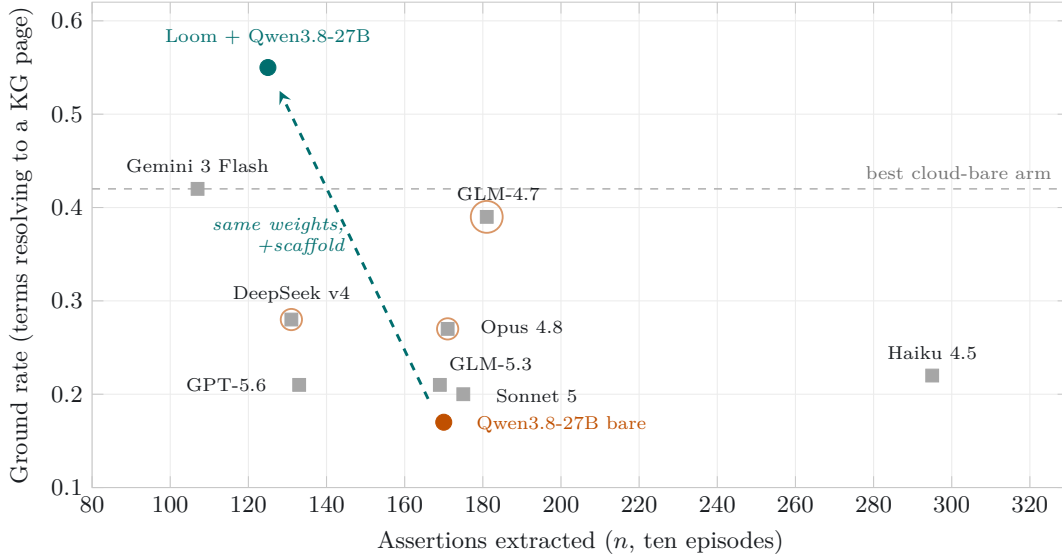


Figure 6: The case study on one panel (Table 8). Each mark is a serving arm: volume of extracted assertions (n) against the rate at which its `ontology_terms` resolve to knowledge-graph pages. Grey squares are cloud models run bare; a burnt-orange halo marks arms with more than two speech-recognition artefacts copied into claims (radius scales with count). The comparison is grounded-local versus bare-cloud, not like-for-like: the dashed teal arrow is the only paired contrast (identical local weights, 0.17 to 0.55 when served through the Loom), while the cloud arms are unaided. Grounded-local lands above every bare-cloud arm here, including the best (dashed grey line, 0.42), but this reflects vocabulary alignment with the graph rather than a head-to-head quality win.

27B (+0.29), DeepSeek v4 (+0.26), Sonnet 5 (+0.25) and Opus 4.8 (+0.20). Three show no measurable composition, but the design bounds what can be detected: since b -only already exposes the target, the composition contrast $\Delta = \text{both} - b\text{-only}$ cannot exceed $1 - b_{\text{only}}$. GPT-5.6’s b -only recall of 0.89, the field’s best guess rate, caps its detectable Δ at 0.11, so its flat result (0.88 both-premises) is consistent with limited headroom rather than evidence the model cannot compose. The per-model b -only rates (guessing baselines, 0.31–0.89) are shown in Figure 7; read each Δ against its own $1 - b_{\text{only}}$ headroom. Four are *anti-composers*, with significantly negative Δ (GPT-4.1-mini -0.47 , Mistral-24B -0.29 , GLM-4.7 -0.18 , Llama-70B -0.12): adding the route to the answer-bearing block reduces performance. Masking relation labels sharply reduces every model (masked recall at most 0.61, most below 0.2, each below its own both-premises rate), so composition depends on labels rather than bare graph structure.

Three conclusions follow. First, the serving boundary is crossable: whereas gain over copy is uniformly negative in the delivery setting, genuine synthesis separates models decisively, and the local model behind this node’s façade sits with the composing class (Gemini 3 Flash’s point delta, +0.33, is nominally larger; per-model intervals overlap and no inter-model significance test is run). Second, guessing from the answer-bearing block alone is substantial (0.31–0.89 recall) and would masquerade as multi-hop reasoning without single-premise controls; uncontrolled two-hop benchmarks therefore overstate composition. Third, 99.97% of naively mined two-hop chains failed the eligibility filters, chiefly because a served block the question

would expose already names the target. With per-IRI serving, most “multi-hop” questions are single-hop exposure in disguise, extending the copy-ceiling lesson one edge outward. Per-model intervals here are individual bootstrap intervals across the twelve-model panel; no multiplicity adjustment is applied to the composer/anti-composer categories.

Lineage and scope. These controls inherit a well-established caution about multi-hop evaluation. The compositionality gap names the tendency of models to answer sub-questions yet fail their composition [49]; MuSiQue builds chains resistant to single-hop shortcuts and adds an unanswerable-decomposition condition [56]; the DiRe probe measures disconnected reasoning directly [55]; and distractibility work shows irrelevant context degrades reasoning even when the answer is present [9, 54]. Our single-premise and distractor arms operationalise these cautions per served block. Two scope notes follow. The anti-composer result is a *deployment-state* observation, not a fixed property: controlled-benchmark work shows such distraction is substantially trainable away [60], so a model that anti-composes today need not after targeted training. And the split by model is consistent with a utilisation threshold: below roughly 7–10B parameters models struggle to use what they retrieve [44], which frames the 27B local model’s place in the composing class rather than explaining it away.

11.1 A second local model family

Findings for the local model above rest on a single candidate, Qwen3.8-27B: it composes (§11), while its

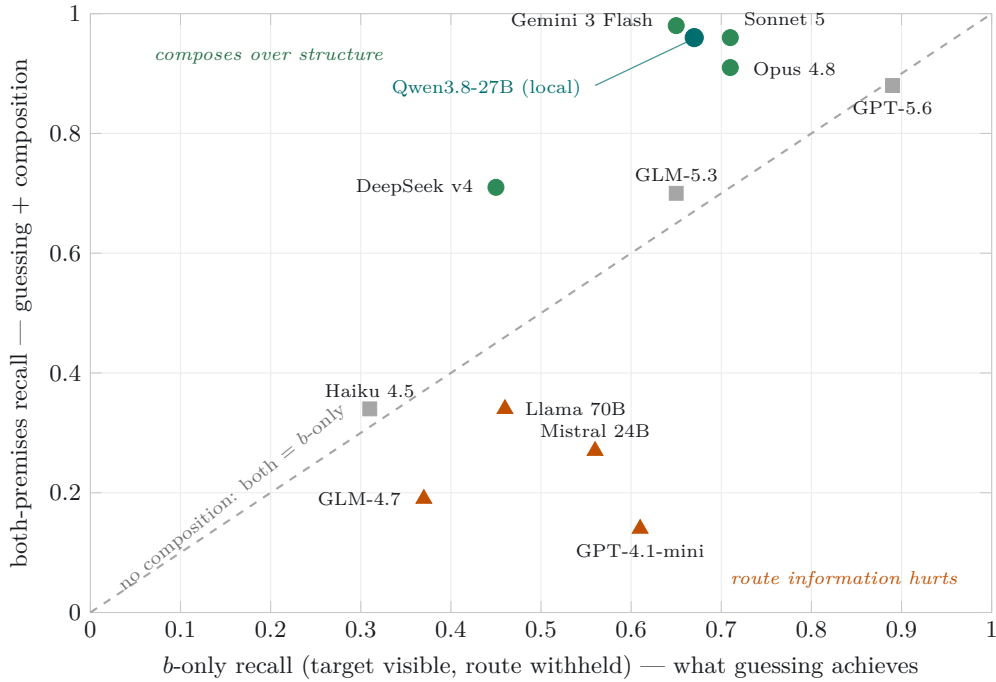


Figure 7: Two-edge composition on one panel. Each mark is a model given the same 150 questions; x is recall when only the answer-bearing premise is served (b -only: the target is verbatim in context but the route from the question’s subject is withheld), y is recall with both premises. The dashed diagonal is the no-composition reference; vertical distance above it is the composition-specific component Δ . Green circles: Δ ’s seeded 10,000-resample bootstrap 95% CI is positive; grey squares: CI spans zero; burnt triangles: CI is negative — for these models the compositional context actively hurts. The three control arms are omitted from the panel because they are uniformly at floor: a -only, matched-irrelevant and raw recall are zero (or $\leq 2/150$) for every model.

delivery saturation is an inference from ceiling headroom rather than a measurement on the 510-question sweep, which does not include it (its Qwen row is Qwen-2.5-72B). Whether these are properties of that one model or of small local models grounded against this corpus is not separable from a single candidate. To test replication beyond it, a second local family, Gemma (gemma-4-31B-it-qat), is run through the two-edge composition test and the Study-1 sweep behind the same façade, holding retrieval, gating, scaffold and decoding fixed and varying only the backend model. This isolates the model contribution to both the composition boundary and delivery-fidelity saturation. Both findings replicate (Table 9). Gemma composes: its two-edge delta is $+0.693$ [$+0.613, +0.767$], above Qwen3.8-27B’s $+0.287$ and every API model measured, with a low b -only recall (0.260) showing it leans on both premises jointly rather than leaking the target from one. And grounded, it saturates the same copy ceiling: raw recall 0.099 lifts to scaffold recall 0.938 against a 0.964 ceiling, a gain over copy of -0.027 [$-0.037, -0.018$] that sits at the least-negative end of the ten-model range and excludes zero. The composer signature and the ceiling-saturation delivery pattern are thus not idiosyncrasies of the first candidate; a second, independently trained local family reproduces both under a single-variable backend swap. Gemma’s sweep run is, moreover, the only local-family delivery measurement on the 510 questions: for Qwen3.8-27B that pattern remains an inference from headroom, as

stated above.

12 A Second Domain: Typed Skill Routing

Every study above measures generation recall on one corpus. Whether the instrument transfers is a separate question, so we applied it to a task of a different shape: a discriminative choice over a curated set, where the model is shown every candidate and must name one, or decline. The first thing it measured was a fault in our own ceiling, and we report that before the result it changed.

Task and exposure. An agent runtime routes each user turn to one of 115 *skills*, each described by a short rubric stating what it is for and when it must not be chosen. The router puts one typed choice to a judge: the turn as state, the 115 rubrics as options, plus a **none** option for turns no skill should handle. The gold label for a turn is the key of one exposed rubric, or **none**. Exposure is therefore total: the gold answer is, by construction, among the text the judge is shown. It is stronger than in Study 1, where 3.6% of gold lay outside the scaffold. We wrote the corpus: 86 labelled turns (16 **none**, 70 skill), each tagged by the class of confusion it was written to probe. The corpus, the three engines and the measurement rig are released

Table 9: Second local family (**gemma-4-31B-it-qat**) through the composition test ($n = 150$) and Study-1 sweep ($n = 510$), same serving path as Qwen3.8-27B, varying only the backend model. Both central local-model findings replicate: a positive composition delta and a uniformly negative gain over copy.

Quantity	Gemma
Composition both-premises recall	0.953
Composition b -only recall	0.260
Composition $\Delta = \text{both} - b\text{-only}$	+0.693 [+0.613, +0.767]
Study-1 raw recall (bare model)	0.099
Study-1 scaffold recall	0.938
Study-1 copy ceiling	0.964
Study-1 gain over copy	-0.027 [-0.037, -0.018]

with the runtime.²

Adapting the ceiling to a choice task. A no-op extractor has no meaning when the answer is a key rather than a passage, so the copy procedure becomes the best *judge-free surface match* over the exposed options: rank the 115 rubrics against the turn and take the top. We use two rankers with no trained judgement in them, BM25 over the rubric text ($k_1=1.5$, $b=0.75$, a fixed stoplist) and cosine similarity under a 384-dimensional sentence embedding. A ranker has no rubric to match a decline against, so conceding **none** to the judge would hand it a large and spurious gain; each ranker is therefore given a decline rule whose threshold is tuned to maximise *its own* accuracy on the evaluated corpus. The ceiling is thereby flattered and the judge’s gain is a floor, provided the tuning finds the maximum it claims to. Ours did not.

What the instrument found first was a fault in itself. Our first implementation reported a lexical ceiling of 76.7% and a judge advantage of +11.6 points. Both figures were artefacts of two defects in the control, each of which inflated the judge:

- 1. The threshold sweep could not find its own optimum.** A “decline below t ” rule is a step function whose only breakpoints are the observed scores, but the sweep evaluated a fixed 41-point grid spanning the observed range. The maximising window here is $9.5715 < t \leq 9.7420$, narrower than one grid step, and the sweep stepped over it: two **none** turns were recoverable at no cost to any correct answer and were never recovered. Correcting the sweep to enumerate every breakpoint moves the ceiling to 79.1% and the gain to +9.3.
- 2. Exclusion clauses were indexed as positive evidence.** Fifty of the 115 rubrics carry a clause saying what the option is *not* for and which option to use instead. Indexed in one undifferentiated bag, those clauses are dense in exactly the vocabulary they disclaim, so a turn about a topic scored *highest*

on the rubric that exists to reject it. Dropping each clause from its marker onward moves the ceiling to 83.7% and the gain to +4.7. The operation uses strictly less of the exposure than the original, so it cannot introduce supervision.

Together the two defects manufactured 8.1 of the original 11.6 points. Through all of it the exposure classification was fixed — which rubric a turn’s gold names was never in question — so the distinction of §3.2 is not abstract here: the counts could not have moved, and the scalar moved seven points. The correction is not a matter of taste: a ceiling that indexes “NOT for X ” as evidence for X , or a tuning procedure that misses the maximum it claims to find, silently withdraws the floor guarantee that is the whole reason for oracle-tuning the baseline in the first place. An instrument that deflates other evaluations has to survive its own test, and ours did not until it was made to.

Result: three judges against one repaired ceiling. Table 10 reports three engines on the same 86 turns against the same two repaired ceilings. They span a wide range of capability: a 421M-parameter encoder whose options share a single 256-token head budget and are each truncated to 48 tokens; a 4B-parameter natural-language-inference cross-encoder scoring each (turn, rubric) pair independently [57]; and the commercial cloud judge whose published figure motivated this deployment.

Against the embedding ceiling both competent judges win decisively and the conclusion does not depend on the instrument chosen. Against the *lexical* ceiling neither does: the cloud judge’s +9.3 points gives an exact p of 0.057 and the local cross-encoder’s +4.7 gives 0.455. The two judges are also not distinguishable from each other (93.0% against 88.4%; 8 discordant turns against 4, exact $p = 0.39$), so the table separates a judge from surface matching only in the semantic case, and does not separate the two judges from each other. Exact power at $n = 86$ against the observed lexical effect is 0.50; roughly 157 turns would be needed for 80%.

Where the difference lies, and where it does not. The aggregate hides the only part of this that

²Sovereign System One, in the agent runtime <https://github.com/DreamLab-AI/agentbox>: a typed-decision wire protocol (choice, score, probability) with a local engine behind it. The router’s request is byte-identical across all three judges below, so the engine is a URL chosen by this measurement.

Table 10: Typed skill routing, $n = 86$, against the repaired ceilings. The ceiling columns are judge-free procedures over the same 115 exposed rubrics, each with an oracle-tuned decline rule, so every gain is a floor. Exact two-sided McNemar on the discordant pairs; the paper’s pre-specified conditional signed-rank test reduces to it exactly on a ± 1 paired difference.

Judge	Top-1	vs lexical ceiling 83.7%		vs embedding ceiling 69.8%	
		Gain	Exact p	Gain	Exact p
421M encoder, shared head budget	53.5	−30.2	—	−16.3	—
4B cross-encoder, independent scoring	88.4	+4.7	0.455	+18.6	0.00086
Cloud judge (jev-1.13.0)	93.0	+9.3	0.057	+23.3	< 0.001

a design choice predicted. Against the repaired lexical ceiling the 4B judge’s advantage is *exactly zero* on ordinary near-neighbour picking (86.0% against 86.0%, $n = 50$) and exactly zero on early-discriminator turns (83.3% against 83.3%, $n = 54$), while all of it sits on boundary cases (+16.7, $n = 12$) and on turns whose discriminating clause falls late in its rubric (+12.5, $n = 32$). That is the correspondence worth reporting: total exposure predicts no gain on ordinary picking, and against a competent ranker one measures none. No subgroup contrast survives Holm correction within its partition, and the two smallest carry four discordant pairs each, where the two-sided exact p floors at 0.125 and no effect size can reach significance — so we report the pattern as a direction the design predicted, not as a set of established contrasts.

Disagreement as a label audit. The ranker and the judge err on different turns, and their union is right on 95.3% against 88.4% for the judge alone. Only 26 of the 86 turns can move a paired comparison at all: the 22 on which the procedures disagree and the 4 both get wrong. A relabelling inside the 60 turns both get right is exactly neutral, since a wrong gold there makes both wrong and leaves the discordant counts untouched. We therefore re-read the complete movable set against the rubrics’ own text rather than only the turns favouring the ranker.

It found one corpus error. For “*Take a screenshot of the staging login page and fill in the demo credentials*” the corpus said **browser**, both judges said **browser-automation**, and the **browser** rubric itself reads “for a browser task choose **browser-automation**, not this”. The judges followed the rubric’s stated boundary; the ranker won on vocabulary overlap. The audit is symmetric by construction rather than by assertion, and the symmetry is testable: turn 28 is the mirror image, where the judge’s pick carries the clause “for a visual critique of an existing UI choose **design-audit**” and the same criterion points *away* from the judge, so the gold was kept; turn 18 is contestable in the judge’s favour and was left standing. Correcting the one label moves the lexical gain by 2.3 points and changes no conclusion in this section, which is the state a single contested label should be in.

A ranker wins by surface, a judge by the rubric’s semantics, so systematic disagreement between them localises either a mislabelled item or a genuinely am-

biguous one. When gold derives from exposed text the copy control is therefore also a pointer at the rows to re-adjudicate — a use the ceiling’s definition does not anticipate, and the one claim here that needs no statistical power and that a stronger ceiling cannot touch. It is also the only finding in this section that survived every correction above unchanged.

What this does and does not show. The instrument transferred to a discriminative task, and its first act was to find two faults in itself that had manufactured two-thirds of the effect it was measuring. What survives is narrow and worth stating exactly: against a semantic ranker a competent judge wins decisively; against a well-built lexical ranker on this corpus it does not win measurably; the difference is located on boundary and late-clause turns and is exactly nil on ordinary picking; and the control found a labelling error its authors had already reviewed. The corpus is the binding limitation: 86 turns, single seed, and prompts we wrote against rubrics we also wrote, whose mean content-token overlap with their own gold rubric is 37.1% against 2.7% for a random rubric. That leakage inflates any surface baseline, so the lexical ceiling reported here is high for a reason particular to this benchmark, and the gain above it is correspondingly a floor rather than an estimate. The method claim is what generalises: a choice task whose gold is among its shown options should report a fair copy ceiling beside accuracy, tune it to a real optimum, and read the gain by population rather than in aggregate.

13 Curated-Corpus Setting

Each design choice measured here follows from a deployment requirement and determines a design outcome; Table 11 shows where each constraint is enforced. The system is a served node³ operating over a corpus produced by a separate, CI-gated pipeline⁴ and accelerated by an in-process HNSW index⁵.

³The served node <https://github.com/DreamLab-AI/loom> (code under the repository licence; corpus terms per the sibling knowledgeGraph repository).

⁴The canonical builder <https://github.com/DreamLab-AI/knowledgeGraph> (published at <https://narrativegoldmine.com>) runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

⁵RuVector: <https://github.com/ruvnet/ruvector>.

13.1 Ecosystem in brief

This report examines one component of the working VisionFlow stack. Knowledge enters through the *knowledgeGraph* repository: 8,138 Logseq markdown pages compiled losslessly into a pure-TBox OWL 2 ontology comprising 8,146 classes and no individuals, then published as versioned generations. Content comes from human-directed authorship and, more recently, automated extraction. Output from OntoCast [19], an independently developed open-source ontology-extraction tool that is not our work, enters through a preview-first import that reports identity collisions, refuses overwrites and emits private review candidates rather than publishing machine output. OntoCast remains an upstream producer at a governed boundary; neither project vendors the other. A CI gate combines consistency tests and a native conflict detector with a *Whelk* OWL 2 EL reasoner [5], classifying each build and rejecting contradictions before publication. The generation includes its reasoned closure of 282,492 triples.

Loom, the measured node, mirrors and serves a generation, combining lexical retrieval over class titles, confidence-gated injection of per-IRI markdown blocks, read-only SPARQL over the closure and delegation to the local model through an OpenAI-compatible façade, currently Qwen3.8-27B on two workstation GPUs. *RuVector* provides corpus embeddings in Postgres and an in-process HNSW semantic fallback. Semantic fallback stays off by default because its measured recall of 0.816 is below the 0.87 design floor. Downstream consumers, including an email gateway and working agents, inherit grounding by holding the façade URL. This context identifies the subject of the deployed-system measurements in §14–§8. Figure 8 shows the full path.

1. **Organisation-controlled-knowledge need $\Rightarrow$ curated, amortised corpus.** The model must answer over a corpus it has not memorised in usable form (raw ≈ 0.26 is consistent with only partial pretraining coverage, not proof of absence; the corpus is public, §1). A human-directed, provenance-tracked ontology incurs curation cost once and is reused per query. The copy ceiling expresses at evaluation time that the curation carries the answer.
2. **Trust $\Rightarrow$ human auditability at single-entity granularity.** A human must be able to audit served knowledge end-to-end. The canonical unit is therefore one per-IRI markdown-with-ontology block, comprising curated prose headed by typed relations, and every retrieval port returns or resolves to an IRI addressing it.
3. **Auditability $\Rightarrow$ accelerators strictly behind the markdown.** The lexical index, HNSW, oxigraph SPARQL and attestation ledger only find, rank or attest the markdown unit and resolve to its IRI. A one-directional module dependency structure enforces this boundary: accelerators may locate the markdown unit but cannot stand in for it.
4. **Privacy / LAN-confinement $\Rightarrow$ local delegation, no cloud on the hot path.** Content and answers remain on the local network; generation is delegated by URL to a LAN/local backend, and augmentation is fully in-process.
5. **Local-model churn $\Rightarrow$ model swap without consumer change.** Local models change often (Gemma $\rightarrow$ Muse-Glimmer $\rightarrow$ Qwen3.8-27B). A stable OpenAI-compatible façade makes the backend one configuration line and records model identity in the result rather than the endpoint, allowing this study to hold grounding constant while varying the model.
6. **Bounded latency $\Rightarrow$ retrieval works with no model, fast.** The scaffold, SPARQL and search endpoints require no model call; the lexical inverted index resolves in **2.02 ms** at the median over 8,146 class titles.
7. **Injection interference $\Rightarrow$ confidence-gated, benchmark-first injection.** Context can displace correct parametric knowledge on off-ontology queries [35, 54, 62]. A single injection authority scales the budget to the retrieval score, skips below a threshold [37], and benchmark-gates features that could add noise.
8. **Never-mixed provenance $\Rightarrow$ attestation and boundary discipline.** Every grounded answer traces to a named, versioned generation and carries a *loom* telemetry block and *corpusNature* provenance metadata. The node serves only the published ontology, never the working graph.

These requirements jointly demand attributable recall where the corpus is authoritative and no regression on general questions the model already answers. We reconcile these axes in §17.1.

14 Ontology Scaffold

Our deployed serving node is a single static Rust binary built as a hexagonal (ports-and-adapters) workspace: a pure domain core defining the port traits, one adapter per backend, a scaffold-policy module, and a thin HTTP façade. This structure enforces the audit boundary of §13: only the domain core mints a served unit, so no adapter can return its own row, triple, or vector as the payload. Golden fixtures pin retrieval, gating, and scaffold serialisation byte-for-byte to a reference implementation.

Canonical unit and the corpus. The unit of service and review is a markdown-with-ontology block scoped to one IRI. Each block contains curated research prose preceded by typed relations: *subclassOf*, *requires*, *enables*, *uses*, *relatedTo*, and *contrastsWith*. The corpus is a single sha-addressable *generation*; the evaluated generation comprises **8,146 concept classes** and **282,492 triples** in the *Whelk*-reasoned closure. Atomic mirroring ensures

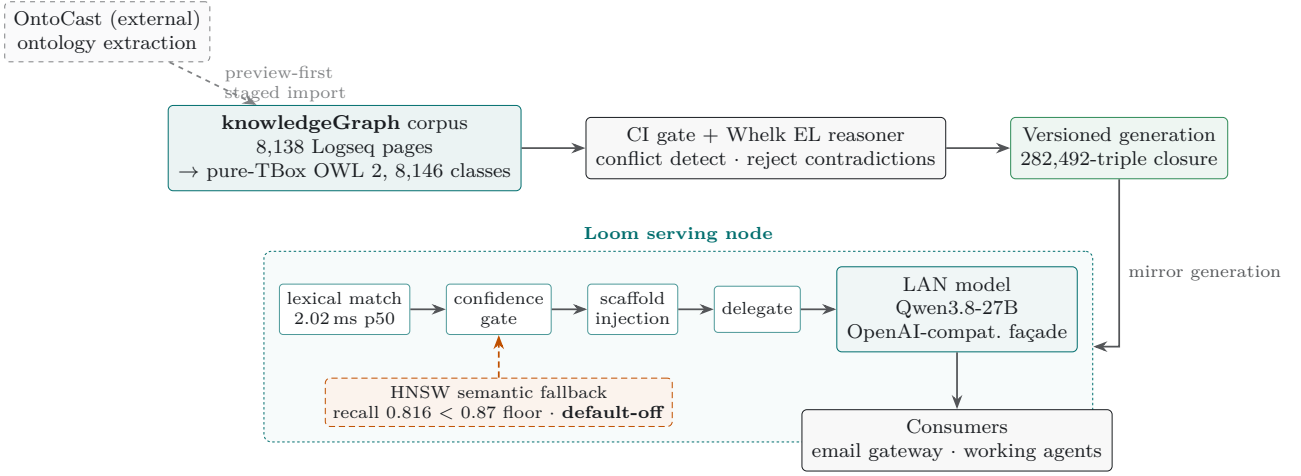


Figure 8: The data path around the measured node. Knowledge enters at the *knowledgeGraph* corpus (8,138 Logseq pages compiled losslessly into a pure-TBox OWL 2 ontology of 8,146 classes); output from OntoCast, an independent third-party extractor, enters only through a preview-first staged import that reports collisions and emits review candidates rather than publishing machine output directly. A CI gate and a Whelk OWL 2 EL reasoner classify each build and reject contradictions before publishing a versioned generation (a 282,492-triple reasoned closure). The *Loom* node mirrors a generation and serves it through a model-free pipeline (lexical match at 2.02 ms, confidence gate, scaffold injection) before delegating to whatever LAN model sits behind its OpenAI-compatible façade (Qwen3.8-27B here); consumers hold the façade URL and inherit grounding. The HNSW semantic fallback is drawn dashed in burnt orange because it ships default-off: its document-embedding recall (0.816) is below the 0.87 design floor, a red gate we report rather than hide (§13.1, §14).

that the node serves a known, mixed-build-free version without embedding data in the binary. Unlike approaches that reduce knowledge to opaque community summaries or GNN-encoded subgraphs [10, 22], this system retains reviewable markdown as the durable asset; the accelerators only locate, rank, or attest to it [17].

Together, the lexical matcher and confidence gate form the exclusive injection authority. A single policy scales the scaffold budget to the retrieval score and injects nothing below a defined threshold, preventing erratic outputs on off-ontology queries (§8.6). The graph store runs read-only, LIMIT-clamped SPARQL over the reasoned closure without a model call. An in-process HNSW fallback over 384-dimensional embeddings supports out-of-vocabulary queries but ships disabled: its document-embedding recall of 0.816 misses the 0.87 design floor. It will remain off until a query-shaped embedding clears a multivariate benchmark, rather than through threshold relaxation; §19 revisits this red gate. Generation is delegated through the model-swap seam to the configured backend URL, currently Qwen3.8-27B. Responses include model identity, while swaps affect no consumer. Thus §7 can vary the model while holding grounding constant, or vary the serving path while holding the model constant. As a deployment consideration, the **2.02 ms** figure is the model-free lexical match alone; a complete grounded answer is dominated by generation on the local backend, with median end-to-end latency of $\approx 90\text{--}170$ s per answer across the Study 2 sets (§8.3).

15 Reasoning-Budget Exhaustion

Serving a reasoning model behind an injected scaffold introduces a measurement hazard that conditions several reporting deviations (§8) and constrains the token budgets used throughout. At `max_tokens=1536`, 42 of 234 loom/raw completions in Study 2 exhausted the budget *during reasoning* and returned no text; the four negative-control arms, left at 1536, lost 36–50% of completions the same way (irrelevant 50.0%, true 42.9%, masked 41.1%, shuffled 35.7%; the ordering does not track scaffold disorder, and no per-length analysis was run, so we report the rates without a monotonic claim). The mechanism is a prompt $\times$ reasoning-budget interaction, with input and output capacity kept distinct: the scaffold consumes the *context window*, while the reasoning trace and the answer consume the configured *completion allowance* (`max_tokens`); a longer or noisier prompt tends to induce a longer reasoning trace, and a reasoning model that exhausts the completion allowance mid-trace emits nothing rather than a truncated answer (backend-specific accounting applies). Two consequences follow for the protocol. First, the completion allowance must cover reasoning-plus-answer, not answer alone; extraction in the case study uses 12,288 tokens and the Study 2 re-runs 4096, with any sub-2000-token allowance treated as unsafe for a reasoning backend behind a non-trivial scaffold. Second, empty completions must be detected explicitly rather than scored as wrong; the common-retry-policy rerun of §8.5 eliminates the attrition (0% at 4096/8192 tokens). Operationally, the first-attempt empty rate was itself a deployment failure signal (42 of 234, 17.9%,

Table 11: Curated-corpus requirements arc: each business requirement, the design consequence it forces, and the point at which the consequence is enforced (not merely intended).

Requirement	Design consequence	Enforcement point
Curated, organisation-controlled corpus	Human-curated ontology, curation amortised once	Sha-addressable generation; raw ≈ 0.26
Human auditability	Per-IRI markdown-with-ontology as the served unit	Ports resolve to an IRI naming one served block
Accelerators behind the unit	Indexes only find/rank/attest the markdown	One-directional module dependencies; accelerators cannot substitute for the unit
LAN confinement	Model delegated by URL; in-process hot path	Backend set by one config value; in-process path is network-free
Model churn	Stable façade; model identity in the result	OpenAI-compatible /v1; one config line
Bounded latency	No-model retrieval path	Lexical index 2.02 ms p50; scaffold needs no model
Injection interference	Confidence-gated selective injection	Single injection authority; skip below threshold
Never-mixed provenance	Per-answer attestation; ontology-only boundary	loom block + corpusNature; working graph never served

Table 12: Request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate reaches the model, whether lexical or, when enabled, semantic.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 $\rightarrow$ seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units’ markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss $\rightarrow$ candidate seeds into the same gate	<i>default-off</i>
Delegate	Inject scaffold into the last user message; call the configured backend URL	yes

in Study 2 at 1536), and complete-case results measure answer quality conditional on an answer being produced, not overall service effectiveness. Temperature 0 also does not guarantee deterministic remote inference; “deterministic” in this paper refers to the scoring and scaffold generation, and to fixed decoding settings, not to provider-side reproducibility. Latency is dominated by generation, not retrieval: the 2.02 ms lexical match is model-free, whereas median end-to-end latency is ≈ 90 –170 s per answer across the Study 2 sets (§8.3).

16 Write-Path Lifecycle

A redesign that replaced inline insertion turns the paper’s instruments into the write path’s architecture. Extracted, verified assertions now land on per-episode *ledger pages* inside the graph⁶, wikilinked to the topics they concern: readers see them immediately through

⁶The ingest pipeline and ledger writer ship in the VisionFlow agent environment: <https://github.com/DreamLab-AI/VisionFlow>; the graph and its CI-gated builder are the knowledgeGraph repository already cited.

linked references, curated prose is never edited, and, because ledger pages carry no ontology markup, nothing enters the served corpus by default. Landing is thereby separated from promotion. A topic whose ledger accumulates evidence, or a thin page whose additions have been judged to improve it, becomes a *candidate*; candidates pass an automated pre-filter consisting of the two instruments this paper developed: the blind before/after quality judgment of §10.1 and the copy-ceiling answer-completeness gate of §9. Survivors then reach the graph’s existing governed proposal queue⁷ as scored dossiers with provenance down to assertion fingerprints. Approval triggers batched section regeneration, which is the point at which content acquires ontology markup and becomes servable. Direct enrichment of thin pages was evaluated as an alternative landing zone and rejected on its judged evidence: its mean stayed negative across twenty pairs

⁷The governed propose/approve write path is the ontology-bridge component of the same stack; the serving façade whose scaffold and copy-ceiling endpoints the pre-filter reuses is the Loom repository already cited.

(−0.65 and −0.60 under two rubrics), and the section-splice mechanic could not apply at all on thirteen of them for want of section anchors, leaving the ledger as the single landing zone. Figure 9 draws the path and its open-source components. The lifecycle applies the paper’s instruments as write-path architecture: the corpus is the asset, every write to it is measured before it is trusted, and the same ceiling that audits the serving layer gates admission to it. Architecturally the ledger-and-served split follows the Event Sourcing / CQRS materialised-view pattern [14] *at the assertion layer*: the ledger is the append-only source of truth for extracted assertions, and the regenerated sections built from it are disposable projections, which is why a diluting projection can be discarded without data loss. Mature curated prose predates the ledger and is retained as a source in its own right; it is never edited on landing and is not rebuilt from the ledger, so the event-sourcing analogy is scoped to the assertion layer, not the whole corpus. Promotion is made idempotent by fingerprinting each assertion and de-duplicating on it, so the at-least-once delivery inherent to such pipelines cannot double-promote.

17 Roadmap and Open Axes

This node clears the parts of the deployment bar it can measure and leaves the rest as declared future work. We record the full bar here as a roadmap rather than a headline claim.

17.1 Multivariate bar

Table 13 specifies the four-axis deployable bar from §13, its instruments and current status. Two axes are measured; two are specified but unresolved. The bar is conjunctive, not reducible to one number.

Axes 1 and 2 establish high-fidelity delivery, carrying per-answer attribution metadata, where the corpus is authoritative, with no detected regression elsewhere; whether that metadata is precise is exactly axis 4, still open. Separate instruments prevent this conjunction from being a single-metric artefact. Axis 3 requires a web-grounded arm on the same questions, since the bare model is not the correct baseline; axis 4 requires reported attribution precision. Both remain unmeasured, so the full bar is not cleared. This qualification prevents the in-domain result from being mistaken for the entire claim, the failure mode that motivates the copy ceiling.

18 Ideal State: A Private Corporate Data Lake

Every study in this paper ran on one deployment, yet the findings constrain a whole class of system: an organisation holding a large private corpus that must be answered from, routed over and kept current, without the content leaving the organisation. This

section states what that system should look like if the measurements above are taken seriously, as a set of properties, since an architecture would over-specify what the evidence supports, and marks which of them the present work has measured and which it has only specified.

Through-line, and why it is the highest-value one. Every property below is a consequence of a single move: **put the same judge-free control at every seam of the lake, so that each decision the system makes about private data is measured against what a verbatim copy of its own input would already achieve, before anyone is asked to trust it.** Promotion of a fact into the corpus, the choice of serving path, the routing of a request, the keeping or dropping of a transcript: each is a place where a model’s output could be mistaken for judgement, and each admits the same per-item, deterministic ceiling and the same signed gain. The justification is empirical rather than aesthetic. Across ten models the gain over copy on the serving path was negative throughout (§8.2), so a lake that pays for synthesis over a curated corpus is paying for reasoning the instrument shows is not occurring; the promotion gate refused a plausible enrichment that a judge would have waved through (§10); and on the routing seam the same instrument said what a judge adds, located it in declining and late-clause discrimination, and flagged corpus errors it had no way to know about (§12). One control, applied everywhere, converts a trust problem into a measurement problem, and the control costs nothing that scales with the model: a copy, a lexical rank, a subtraction. That is why it outranks the other six properties. Curation, egress control and reversible writes are ordinary good practice that any organisation can adopt without this paper; the ceiling is what makes them checkable, and it is the one property that tells the organisation where its money is and is not doing work.

Properties of the ideal state.

- D1 A curated core with a promotion gate.** (The through-line applied to admitting facts.) The corpus is the curated, provenance-carrying subset of the lake that the organisation is prepared to serve as fact; the rest of the lake is raw material. Candidates enter through a gate that computes the copy ceiling before promotion, so an extraction whose recall does not exceed what its own source already exposes is rejected as restatement (§10). *Measured*: the podcast assertion pipeline and its judged-page gate.
- D2 A delivery-first serving path.** (The through-line applied to answering.) The default answer path is retrieval plus a static scaffold, and its quality is reported as gain over copy by population rather than as raw recall. Synthesis machinery is admitted only where that gain is positive and located, since the alternative is paying for

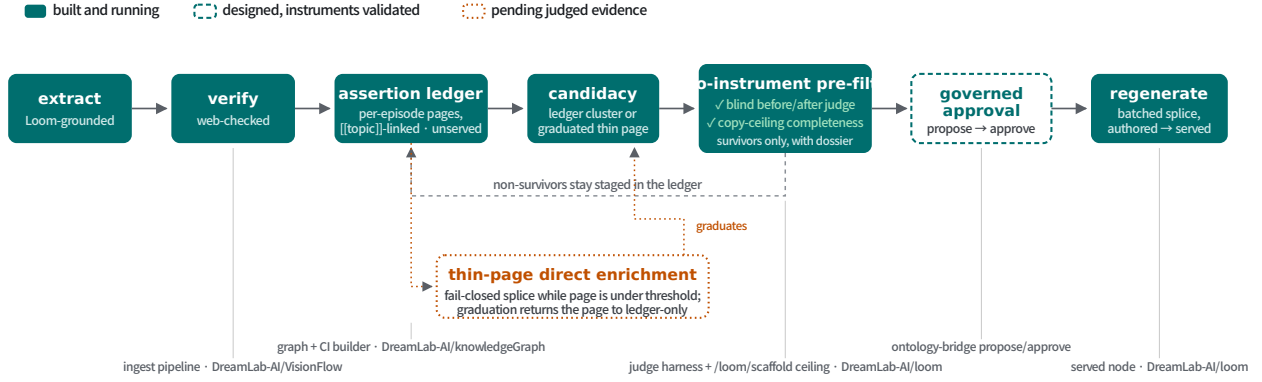


Figure 9: Write-path lifecycle and the open-source component carrying each stage. Landing is separated from serving: assertions stage on unserved ledger pages, candidates pass the two instruments this paper develops, and only governed, regenerated content enters the served corpus. The thin-page branch is conditional on its own judged evidence and self-terminates: an enriched page that crosses the thinness threshold returns to ledger-only treatment.

Table 13: Multivariate bar and this work’s status. A deployable curated-knowledge system must clear all four *together*; the axes are measured on different instruments, so no single number establishes the conjunction.

Axis	Instrument & evidence	Status
1. In-domain delivery fidelity	Copy ceiling / gain over copy: ten models -0.067 to -0.022 ; live loom—raw $+0.27$ pooled, $+0.79$ arcane	Measured (both studies)
2. Out-of-domain non-regression	Judged Δ vs frontier gold: five-model arm all CIs include zero (worst -0.23 $[-0.63, +0.05]$, wider than the margin, so equivalence not established there); live general $+0.05$ $[0.00, +0.13]$ inside the ± 0.25 margin by interval inclusion	Measured (both studies)
3. Beats web search in-domain	Same model grounded in live web results vs curated scaffold, identical question set	Specified; not measured
4. Attribution precision	Per-answer generation + confidence already emitted; precision metric to close “faithful” $\rightarrow$ “attributable”	Specified; not measured

reasoning that the ceiling shows is not happening (§8.2, §8.5). Two measured facts bound this property and belong in any deployment decision. Grounding has a cost: on the items the corpus does not expose, recovery falls from 0.121 bare to 0.004 grounded (§9.1), so a delivery-first path makes the model roughly thirty times worse at anything outside the corpus, and the lake must route out-of-corpus questions away from the scaffold rather than through it. And the specificity of retrieval is unproven: the control rerun shows the scaffold beats no context but does not establish that the retrieved content beats generic on-corpus text of the same length (§8.5), so the value demonstrated so far is that of on-corpus text being present, not of the retriever choosing well. *Measured*: in-domain delivery fidelity; out-of-domain non-regression. *Open*: the web-search baseline and attribution precision (§17.1).

D3 Typed decisions at every seam, on local judges. (The through-line applied to deciding.) Routing, keep-or-drop, verification and re-ranking are small typed judgements, answered by a judge that scores options independently so that declining

is a threshold rather than a competitor, and each judgement is shipped with a fair copy ceiling and read by population (§12). A judge-free ranker answers first and the judge is called on escalation; on the routing corpus that cascade matches the judge at two-fifths of its latency. The implementation is the typed-decision façade of §12: consumers speak one wire protocol (a choice, an ordered score or a probability), and the judge behind it is whichever engine last won the eval. *Measured*: routing, on one small corpus, through that façade. *Specified*: keep-or-drop over transcripts, where the same instrument applies and the decline asymmetry is sharper; and the served node’s own promotion and judged-page gates, which still run on prosemitting LLM judges and are the next seam to move behind the façade.

D4 A hard egress boundary with a swappable model behind a façade. No corpus content, transcript or judgement state leaves the boundary; the model is a URL behind a stable façade and is replaced on the strength of a measurement, with no change on the consumer side. The routing replication was itself such a swap: the failing en-

gine and its successor sat behind one wire protocol, and the eval decided between them where a vendor’s claim could not have. *Measured*: the swap and its cost; *specified*: the boundary as policy.

D5 Standing instrumentation, not one-off evaluation. (The through-line as an operating discipline.) Every served path reports its ceiling, its signed gain, its population breakdown and its power, as a matter of course; the ten-model and routing tables are what that reporting looks like. Where a ranker and a judge disagree, the items are queued for label audit rather than counted silently against either (§12). *Measured*: the instrument; *specified*: its standing operation.

D6 A gated, reversible write path. Additions to the corpus carry provenance, pass the promotion gate, and can be withdrawn; the lifecycle in §16 is the shape, and thin-page enrichment is the example of a plausible addition that the gate correctly refused. *Measured*: the refusal; *specified*: reversibility at scale.

D7 Stated limits. The system delivers vetted knowledge faithfully and attributably at low cost. Discovery is outside its remit and should stay outside its marketing: on unexposed items the grounded path recovers less than the bare model does. A lake that promises reasoning over private data is promising the thing the ceiling shows is absent.

What aggregate means here. None of the seven properties is novel on its own; several are ordinary data-governance practice. The claim is about their composition: that a private data lake whose every seam is a typed judgement with a copy ceiling beside it, whose serving path is delivery-first by measurement, and whose corpus admits facts only past that same ceiling, is a system whose behaviour can be stated and checked, which is a stronger position than one that has to be trusted. The present work has measured the instrument, the serving path on one corpus, one promotion gate, one model swap and one routing seam. It has not run the seven properties together as one deployment, and that composition is the next thing to measure, with the same discipline: ceiling first, gain by population, power stated, and the model behind a door.

19 Limitations and Threats to Validity

- **Recall rewards a verbose copier.** The Study-1 metric is recall against graph-derived gold, so a model that emitted the entire scaffold verbatim, gold together with arbitrary fabrications, would score at the copy ceiling. Precision and attribution are consequently unmeasured, and “faithful delivery” here means high recall of exposed gold, not verified absence of fabrication. The multivariate

bar (§17.1) is accordingly a framework of which only two of four axes are closed (in-domain delivery fidelity and out-of-domain non-regression); the precision/attribution axis is specified but open.

- **Single corpus.** Both studies use one curated ontology in one broad domain. Although the copy ceiling and decomposition are corpus-general, their magnitudes (a 0.964 ceiling and +0.27 live lift) do not transfer.
- **Judge coverage.** Study 2’s headline uses one GPT-family judge (`openai/gpt-4.1`), cross-family from the Qwen candidate, which limits self-preference; a full third-family re-judge (`claude-opus-4-6`, §8.4) reproduces the contrast. There is still no per-answer multi-judge panel, answer-order randomisation, or repeatability test across judge seeds, and gold references mitigate but do not eliminate judge idiosyncrasy.
- **Control rerun: what it settles and its own caveats.** The uniform-budget rerun (§8.5) removed the 36–50% attrition, added a length-matched fluent-noise floor, and applied Holm correction; it downgrades the content-specificity claim (only true—raw and shuffled—raw clear correction) rather than confirming it. The rerun carries its own limits: it re-judged with `gemini-3.1-pro-preview` (forced by exhausted `gpt-4.1` credits), so the control judge differs from Study 2’s `gpt-4.1` headline: old and new control rows share the Gemini judge, isolating the budget effect, but cross-study judge comparisons should not be made; per-arm n is still modest (≈ 50 –55); and both studies use a single local production model.
- **Small per-set n for controls.** Per-set contrasts have $n = 8$ –21. The curation-depth gradient (arcanic true—raw +0.95 significant, thin +0.08 ns) is suggestive, while thin-set nulls may reflect either low power or no effect.
- **Lexical scorer for Study 1, checked semantically.** The sweep’s lexical title matcher undercounts paraphrase, making absolute recall a lower bound; a paraphrase-tolerant re-score confirms the negative gain is not an artefact of strict matching (§4).
- **Semantic fallback disabled throughout, now characterised.** HNSW semantic retrieval was disabled because its document-embedding recall (0.816) falls below the 0.87 design floor, a red gate retained and reported; the loom arm uses lexical retrieval only. §6 now measures what the gate is worth: on vocabulary-mismatched queries the semantic path carries real recoverable exposure, but the shipped absence-keyed trigger cannot reach it.
- **Single production model.** Study 2 fixes the model at Qwen3.8-27B, isolating the serving-path effect but leaving its magnitude under other backends unmeasured. The ten-model sweep varies models only on the offline scaffold, not the live node.

- **Three excluded questions.** Three thin-set questions still empty in one arm at 4096 are excluded pairwise (complete-case $n = 114$). The exclusion rule is symmetric, but the realised missingness was not (loom empty once, raw twice), and complete-case analysis departs from intention-to-treat; a deployment-effectiveness claim would need the empty answers scored as failures.
- **Local-model identity.** The local backend is addressed by a compatibility alias (`qwen3.8-27B`); the deployed checkpoint is a quantised derivative recorded in the node’s deployment notes rather than an attested reference release, and provider display names elsewhere in the paper are likewise not immutable version identifiers. Exact request-level provenance (checkpoint revision, quantisation, server version, run dates) belongs to the release manifest rather than this text.
- **Two axes of the bar unmeasured.** The specified web-search baseline (axis 3) and attribution precision (axis 4) remain unmeasured, so the deployable claim is not fully closed.
- **The delivery studies cannot license reasoning claims.** Most in-domain targets (96.4%; 76 per model are not) are verbatim in the scaffold, and beyond-exposure recovery is negligible ($n_{01} = 3$ of 11,360). The 11,360 figure pools the same 1,136 target instances across ten models, not 11,360 independent facts. The multi-hop design separating composition from exposure and guessing is reported in §11; its limits (two relation types, one corpus, a lexical scorer, single run per arm) remain those of a single-configuration case study. The same caveat applies across the paper: questions are sampled under a single seed (42); the bootstrap intervals estimate uncertainty under resampling of the frozen questions, but do not cover corpus, snapshot, prompt or seed variation; a seed sweep is future work.
- **The scalar is implementation-sensitive where the counts are not.** A copy ceiling for a task with a natural no-op extractor (Study 1) is a matcher applied to fixed text. For a task without one (§12) it must be constructed from a ranker, a decline rule and a segmentation, and each is a choice: one definition implemented three ways moved the gain by seven points. We report the scalar because it is quotable and comparable across models, but a reader of a constructed ceiling should ask what it was built from, and the counts of §3.2 are the figures to compare across papers.
- **The routing replication is a method transfer, not a powered result.** §12 runs the ceiling on 86 turns we wrote against rubrics we also wrote, so the phrasing echoes them: mean content-token overlap with the gold rubric is 37.1% against 2.7% for a random one, which inflates any surface baseline and makes the reported gain a floor rather than an estimate. Exact power at $n = 86$ is 0.50 and roughly

157 turns would be needed for 80%, so only the embedding contrast is adequately powered. The copy procedure is an adaptation (a ranker with an oracle-tuned decline rule) rather than a verbatim extractor, its exclusion-clause segmentation is a choice we made and report, and the late-discriminator tag depends on a 48-token cut inherited from a superseeded engine. The three judges differ in training data as well as architecture, so the table is not a controlled architecture comparison.

20 Conclusion

Grounding evaluations where the gold standard derives from the injected corpus should account for exposure: classify each gold item by whether it was shown and whether it was recovered, report the four counts, and quote the copy ceiling as their per-item scalar. The counts distinguish dropped exposed facts from beyond-exposure recovery, which neither a bare-model baseline nor the scalar alone can do, and they depend on nothing but a matcher and fixed text. The scalar is the number a reader quotes; on our own second domain it moved seven points under repairs to one definition while the counts could not move at all. Across ten models, gain is uniformly negative. In a second domain, a discriminative routing task with total exposure, the same instrument first located two faults in our own ceiling and then, repaired, returned a positive gain against a semantic ranker, none distinguishable against a lexical one, and a difference concentrated on boundary and late-clause turns rather than on ordinary picking (§12); an instrument that deflates other evaluations must survive its own test first. Of 11,360 gold items, only 3 unexposed items are recovered; context utilisation is 0.900–0.955, so beyond-exposure recovery is negligible. A production paired study lifts judged quality by +0.27 pooled and +0.79 where curation is deep. A common-retry-policy, single-judge control rerun shows the scaffold as a whole reliably beats no context, but does not establish that the specific retrieved content beats generic on-corpus text of the same length; the control design earns its keep by withholding that stronger claim, not by granting it. The reusable method is *the ceiling plus a verified-irrelevant placebo*, computed before architecting to distinguish a serving regime, where verbatim retrieval wins, from a synthesis regime, where heavier machinery must be measured above the ceiling rather than the bare model. Value therefore concentrates upstream at curation, where the instrument also serves as a promotion gate for extraction candidates. For a curated corpus, the goal was never a model that reasons its way to answers it never saw. It is a model that delivers vetted knowledge faithfully and attributably at low cost, and an instrument that measures whether it does.

A Judge rubrics (excerpted)

For reproducibility we reproduce the judge prompts as run, with elisions explicitly marked: the answer judge below is complete; page rubric A elides only the inserted data blocks (marked [...]), and rubric B is reproduced as its differences from rubric A rather than in full. The complete templates with marked data slots ship with the released harness. The production-node semantic answer judge (§7, behind the +0.27 headline) grades a candidate answer against a reference on a 0–5 scale:

You are a strict, fair evaluation judge. Grade the CANDIDATE ANSWER against the REFERENCE ANSWER for the QUESTION, on a 0–5 integer scale:
5 = fully correct and complete vs the reference; 4 = correct, minor omission;
3 = partially correct, no major error; 2 = mostly wrong or one right point;
1 = on-topic but no correct content; 0 = wrong, irrelevant, or fabricated.
Judge CONTENT match to the reference, not style or verbosity. If the candidate adds correct information beyond the reference, do not penalise it. If it contradicts the reference, penalise heavily.
Respond with ONLY a JSON object: {"score": <0–5>, "why": "<one sentence>"}.
[... Page topic, VERSION A, VERSION B blocks ...]

The judged-page gate (§10.1) scores a blind before/after page pair. Rubric A is the original quality rubric; rubric B was added post-hoc, disclosed as such, to credit informativeness after rubric A returned a uniformly negative result across every arm; both are reported side by side and rubric B validates rather than replaces the rubric A measurement.

Rubric A (original quality):

You are evaluating two versions of a knowledge-base wiki page on a technology/AI topic. You do not know how these versions were produced or what process created them – just read them as a reader would and judge quality.
Score VERSION B RELATIVE TO VERSION A on this rubric. For each numeric field, score VERSION B's absolute quality (0–5, 5 best); "improvement" should reflect whether B is better or worse than A specifically.
Return STRICT JSON only:
{
 "factual_grounding": <0–5, VERSION B's factual grounding/specificity>,
 "relevance": <0–5, VERSION B's relevance/focus on the page topic>,
 "coherence": <0–5, VERSION B's internal coherence and readability>,
 "better_version": "A" | "B" | "tie",
 "improvement": <-2..2, how much better(+)/worse(-) B is than A overall>
}

Rubric B (post-hoc, informativeness-aware) differs only in the framing and the relevance/improvement fields:
A reader consults this page to get current, accurate knowledge of the topic.
Which version better serves that reader? Weigh informativeness and currency of content alongside prose quality – a page that omits significant recent developments serves the reader worse, and new content is valuable when accurate and relevant, though it must still be well-integrated.
... "relevance" and "improvement" are scored for a reader seeking current, accurate knowledge; the JSON shape is otherwise identical to rubric A.

B Worked Example: Question, Gold, Paraphrase and Scaffold

A single item from the sweep makes the copy-ceiling inputs concrete. The question, its gold target set, and the paraphrase are all verbatim from the frozen sets; the paraphrase is the stored vocabulary-mismatch rewrite (§6), which preserves meaning while excluding the seed class's title tokens and resolves to the correct class for at least one model family (resolvable_any = true).

QUESTION (id q0003):

In the DreamLab knowledge graph, what is the immediate parent concept of Affective Computing? Also name up to three broader ancestor concepts.

GOLD (slug/title targets asserted by the graph):

{ human-computer-interaction / "Human Computer Interaction" }

PARAPHRASE (stored rewrite; seed title "Affective Computing" excluded):

Within the DreamLab schema, what is the direct taxonomic predecessor of the domain involving machines that process human emotions, and what are three of its more distant lineage categories?

The deterministic scaffold served for this question (budget 1500 tokens, max_seeds 4, hops 1, prose off) begins:

The following ontology context was retrieved from a curated knowledge graph.
Where it is relevant to the user's request, treat it as ground truth for

definitions and relationships between the concepts it covers. Where it is not relevant, ignore it and answer normally.

[ONTOLOGY CONTEXT]

Affective Computing (ai, maturity: emerging)

Affective computing is a branch of artificial intelligence and human-computer interaction concerned with systems that can recognise, interpret, process, and simulate human emotions and affective states. [...curated prose continues...]

is-a: Human Computer Interaction; ancestors: AI Research Area,
Artificial Intelligence

relations: requires: Emotion Recognition, Annotated Training Data, ...

Here the gold title “Human Computer Interaction” appears verbatim in the scaffold (as the **is-a** parent), so the per-item copy ceiling is $c_i = 1$: a no-op extractor of the context would already score the answer. This is the serving-regime signature the ceiling detects. Two properties of this frozen item need stating so the example is not over-read. The question template asks for broader ancestors as well, but the graph asserts gold for this item as the immediate parent alone (T-TAX gold sets in the frozen corpus range from one to three targets, depending on what the graph asserts for the seed class); the ancestor names visible in the scaffold are context, not gold. And although the engine ran with **prose off**, each served block always carries its curated definition text — the flag controls additional generated prose, not the block’s own content.

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